

Propagation of meteorological drought to streamflow and baseflow droughts using an improved SWAT model

Lei Jin^{a,b}, Huazhu Xue^c, Guangyu Ma^d, Yinglan A^{a,*}, Guoqiang Wang^{a,*}, Kaiji Li^{a,b}, Jin Wu^a, Yuntao Wang^a, Baolin Xue^a

^a Advanced Interdisciplinary Institute of Satellite Applications, Beijing Normal University, Beijing 100875, China

^b Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China

^c School of Surveying and Land Information Engineering, Henan Polytechnic University, Jiaozuo 454000, China

^d Environmental Development Center of the Ministry of Ecology and Environment, Beijing 100029, China

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ABSTRACT

Study region: The Jing River Basin (JRB) and Tao River Basin (TRB) on the Loess Plateau (LP), China.

Study focus: Baseflow plays a critical role in hydrological drought evolution, yet baseflow drought and its propagation processes remain poorly understood. Moreover, hydrological drought assessments are often constrained by low simulation accuracy and considerable uncertainty. To address these limitations, this study applies an improved Soil and Water Assessment Tool (SWAT) model in combination with the Eckhardt digital filter and run theory to simulate streamflow, separate baseflow, and identify drought events. The characteristics of meteorological, streamflow, and baseflow droughts, together with their propagation dynamics, are systematically analyzed and compared.

New hydrological insights for the region: The improved SWAT significantly enhances streamflow simulation accuracy, providing a reliable basis for drought analysis. Meteorological droughts occur more frequently than hydrological droughts, whereas hydrological droughts, particularly baseflow droughts, exhibit longer duration and greater severity. Propagation from meteorological drought to streamflow drought shows higher effective propagation rates and shorter response times than propagation to baseflow drought, indicating a more sensitive response of streamflow to precipitation variability. Basin comparison reveals faster propagation to streamflow drought in the TRB than in the JRB, while propagation to baseflow drought is slower and shows contrasting temporal patterns. These findings highlight the importance of baseflow drought in hydrological drought analysis and support watershed scale water resource management.

1. Introduction

Global climate change represents one of the most pressing environmental challenges confronting human society. Ongoing global warming has intensified the occurrence of extreme climate events, among which drought remains one of the most widespread and impactful natural hazards (Han et al., 2023; Wan et al., 2023). Recent studies indicate that rising global temperatures contribute to

* Corresponding authors.

E-mail addresses: ayl@bnu.edu.cn (Y. A.), wangggq@bnu.edu.cn (G. Wang).

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more persistent drought conditions, especially in semi-arid and arid regions, where drought impacts have become increasingly severe (Wang et al., 2021b; Wu et al., 2024a). Drought not only adversely affects global plant growth and agricultural production but also undermines the long-term sustainability of water resources (Pan et al., 2023; Xue et al., 2024). Based on the affected areas and the manifestation of water shortages, drought can be categorized as meteorological, hydrological, agricultural, and socio-economic (Mishra and Singh, 2010). Different types of droughts can occur independently or sequentially, with one type propagating through the hydrological cycle to another, a process known as drought propagation (Gevaert et al., 2018; Lin et al., 2023). In-depth investigation of the characteristics of different types of droughts and their propagation patterns helps to comprehensively understand the evolution mechanisms of drought and provides more accurate scientific foundation for water resource management.

Recent studies have substantially advanced the understanding of drought propagation, with particular emphasis on the transition from meteorological drought to agricultural and hydrological droughts (Wang et al., 2025; Zhu et al., 2021) (Ding et al., 2021b; Gu et al., 2020). In the study of meteorological-hydrological drought propagation, hydrological drought is often characterized by generating streamflow drought indices based on streamflow observations collected at hydrological stations (Wang et al., 2023; Yu et al., 2023). However, limitations arising from the uneven spatial distribution of monitoring stations, especially the scarcity of hydrological stations in remote mountainous areas, hinder accurate representation of drought conditions across different areas of a watershed (Lai et al., 2019). In this context, streamflow data simulated by hydrological models becomes particularly important, as it facilitates deeper insights into the features of hydrological drought across different regions and the physical processes governing drought propagation, thus compensating for the spatial coverage limitations of traditional monitoring data (Pan et al., 2020; Yang et al., 2024). As an effective hydrological simulation tool, the SWAT model is frequently applied in drought research, especially in the study of basin-scale hydrological processes and drought propagation (Dash et al., 2019; Kang and Sridhar, 2017). For example, Chen et al. (2024) used the SWAT to study the integrated propagation characteristics of meteorological and hydrological droughts at the scale of hydrological response units and explored the influence of different land surfaces on drought propagation. Zou et al. (2024) integrated the SWAT and Bayesian networks to explore spatiotemporal patterns of meteorological and hydrological droughts under changing climate conditions, emphasizing the role of climate factors in the drought propagation process.

Although hydrological models can elucidate the physical processes underlying drought propagation, they require highly accurate input data and involve a certain level of uncertainty (Zhou et al., 2024). By improving the hydrological model, the precision and trustworthiness of the simulation outputs can be enhanced to a certain extent. Currently, significant progress has been made in the research on improving the SWAT model, primarily focusing on enhancing the precision of model parameterization, optimizing input data processing methods, and improving the model's adaptability. (Jiang et al., 2024; Lai et al., 2020; Pang et al., 2020). For example, Chen et al. (2023) corrected the vegetation growing season of the SWAT using remote sensing phenological data, improving the accuracy of the model in simulating streamflow and evapotranspiration. Jiang et al. (2024) downscaled precipitation data using the WRF and integrated higher-quality precipitation datasets into the SWAT, significantly enhancing the model's performance in simulating streamflow. These improvements help to increase the model's adaptability across various geographical regions and climate conditions, thereby further enhancing reliability of hydrological drought forecasts and the analysis of propagation patterns.

Additionally, most studies of hydrological drought concentrate on streamflow drought, utilizing variations in streamflow to evaluate drought occurrence, progression, and impacts (Huang et al., 2017; Zhang et al., 2022). In contrast, baseflow drought has received much less attention, despite the critical role of baseflow in maintaining river discharge and ecosystem stability during dry periods (Miller et al., 2016; Yao et al., 2017). Because baseflow drought is more directly controlled by subsurface storage, groundwater recharge, and delayed catchment response, it may exhibit longer propagation lags, longer durations, and greater severity than streamflow drought under prolonged precipitation deficits (Wang et al., 2022). Although recent studies have increasingly emphasized the importance of baseflow drought for drought recovery and long-term water security, research on the propagation from meteorological drought to baseflow drought remains limited, particularly at the watershed scale and in ecologically fragile regions (Ghaneai et al., 2026; Xiong et al., 2025). For example, Lee and Ajami (2023) analyzed 358 watersheds across the United States and found that baseflow drought can be more severe than meteorological drought and may require a longer recovery time. This finding highlights the need to further clarify the propagation mechanisms of baseflow drought.

The Loess Plateau is a typical ecologically fragile region characterized by severe soil erosion, limited water availability, and constrained groundwater recharge, which make hydrological processes highly sensitive to climatic variability and drought (Jia et al., 2024; Qiu et al., 2024). The Jing River Basin (JRB) and Tao River Basin (TRB) were selected as representative basins of the Loess Plateau because both are drought-prone but differ in hydroclimatic conditions and hydrological responses, providing a useful basis for comparative analysis of meteorological, streamflow, and baseflow drought propagation. Against this background, this study aims to: (1) improve the SWAT model and evaluate the performance of the improved SWAT in simulating streamflow in different watersheds; (2) analyze the spatial distribution characteristics of streamflow and baseflow based on the improved SWAT and baseflow separation method; and (3) investigate the characteristics, propagation patterns, and spatial heterogeneity of meteorological, streamflow, and baseflow droughts using run theory and the maximum correlation coefficient method. By analyzing the simulated hydrological results of these two watersheds, this study is expected to improve understanding of the propagation of meteorological drought to streamflow and baseflow droughts, particularly the distinct response of baseflow drought, and to demonstrate the value of the improved SWAT for drought propagation analysis.

2. Materials and methods

2.1. Study area

The Loess Plateau (LP), an important natural geographic unit of China, is characterized by significant topographic variation, with a typical plateau-hill landform that gradually descends from the northwest to the southeast (Jiang et al., 2017). The Jing River Basin (JRB) and Tao River Basin (TRB), located on the LP, were selected as the study area (Fig. 1), representing an arid–semi-arid transition zone. (Wu et al., 2024b). The JRB, situated in central Shaanxi Province, is a representative basin of the Loess Plateau, covering approximately $1.4 \times 10^4 \text{ km}^2$. In contrast, the TRB is located in central Gansu Province and spans about $2.4 \times 10^4 \text{ km}^2$. The Tao River is a tributary of the Yellow River system, and its basin exhibits pronounced topographic variability, characterized by alternating distributions of mountains, hills, and basins.

The region has distinct climatic characteristics, belonging to a temperate continental monsoon climate (Han et al., 2021). Annual precipitation is generally low, with rainfall mainly concentrated during the summer flood season, while winters are relatively dry, leading to scarce water resources and severe soil erosion in the area (Liu et al., 2024). The distinctive topography and climatic conditions of the LP make drought events in the JRB and TRB particularly pronounced, exerting substantial impacts on agriculture, water resources management, and ecosystem stability. Investigating the spatiotemporal dynamics of meteorological, streamflow, and baseflow droughts, together with their propagation processes within these basins, provides essential scientific insights to support regional drought mitigation and water resources planning.

2.2. Data description and processing

2.2.1. Topographic, land use, and soil datasets

This study employs multiple spatial datasets, including a digital elevation model (DEM), land use information, and soil properties. The DEM provides topographic information of the study area and is primarily used for watershed and sub-watershed delineation, river network generation, and the extraction of topographic factors such as slope. The DEM used in this study has a spatial resolution of $30 \text{ m} \times 30 \text{ m}$ and was sourced from the Geospatial Data Cloud (<https://www.gscloud.cn/>). Land use data were utilized to delineate distinct land use units, which were then integrated with soil type and slope information to define Hydrological Response Units (HRUs) within the SWAT. The land use data were obtained from the CLCD dataset (<https://zenodo.org/record/5210928>) with a spatial resolution of $30 \text{ m} \times 30 \text{ m}$, and the land use map for the year 2010 was used for SWAT model construction. Soil datasets characterize soil types and properties, serving as key inputs for the simulation of infiltration, evaporation, and streamflow. The soil data were obtained from HWSO version 1.2, with a resolution of $1 \text{ km} \times 1 \text{ km}$.

Based on the above data, ArcGIS 10.6 software was used for spatial clipping, and the ArcSWAT module was employed to define and delineate the watershed. As a result, the Jing River basin was divided into 47 sub-watersheds and 738 HRUs, while the Tao River basin was divided into 58 sub-watersheds and 796 HRUs.

2.2.2. Observed streamflow and meteorological data

Meteorological forcing data were derived from the CN05.1 dataset (<http://data.cma.cn/>), including daily precipitation, maximum and minimum air temperature, wind speed, and relative humidity, at a spatial resolution of 0.25° and a time span from 2001 to 2020 (Wu and Gao, 2013). Based on the acquired meteorological and soil data, the required meteorological and soil databases for the SWAT were generated to facilitate model construction. The streamflow data were sourced from the Gansu Provincial Department of Water Resources (<https://slt.gansu.gov.cn/>), covering monthly streamflow data for the JRB and TRB from 2003 to 2020.

2.2.3. MODIS data

The MODIS data used in this study include MOD15A2H LAI data and MCD12Q2 vegetation phenology data, both with a spatial resolution of 500 m and temporal resolutions of 8 days and 1 year, respectively. These datasets were obtained from the Google Earth Engine cloud platform. Due to interference from clouds and precipitation, the LAI (Leaf Area Index) data contain missing values and anomalies. To obtain continuous and high-quality LAI data, this study employed the iterative Savitzky–Golay filtering method for data reconstruction. This method effectively smooths the data, removes noise, and fills missing values, thereby improving data continuity and accuracy. The specific implementation process can be referenced in relevant literature (Chen et al., 2004).

2.3. Methods

2.3.1. Overview of the SWAT model

The Soil and Water Assessment Tool (SWAT) is used to simulate hydrological processes at the watershed scale, including precipitation–runoff processes, soil moisture dynamics, and water quality components. The model has been widely applied in watershed management and water resources assessment (Aloui et al., 2023; Zhao et al., 2024). SWAT simulations are based on a water balance framework, in which hydrological components are calculated independently for each HRU, and streamflow is subsequently routed through the river network.

For streamflow simulation, SWAT commonly adopts the SCS-CN method (Eqs. (1) and (2)):

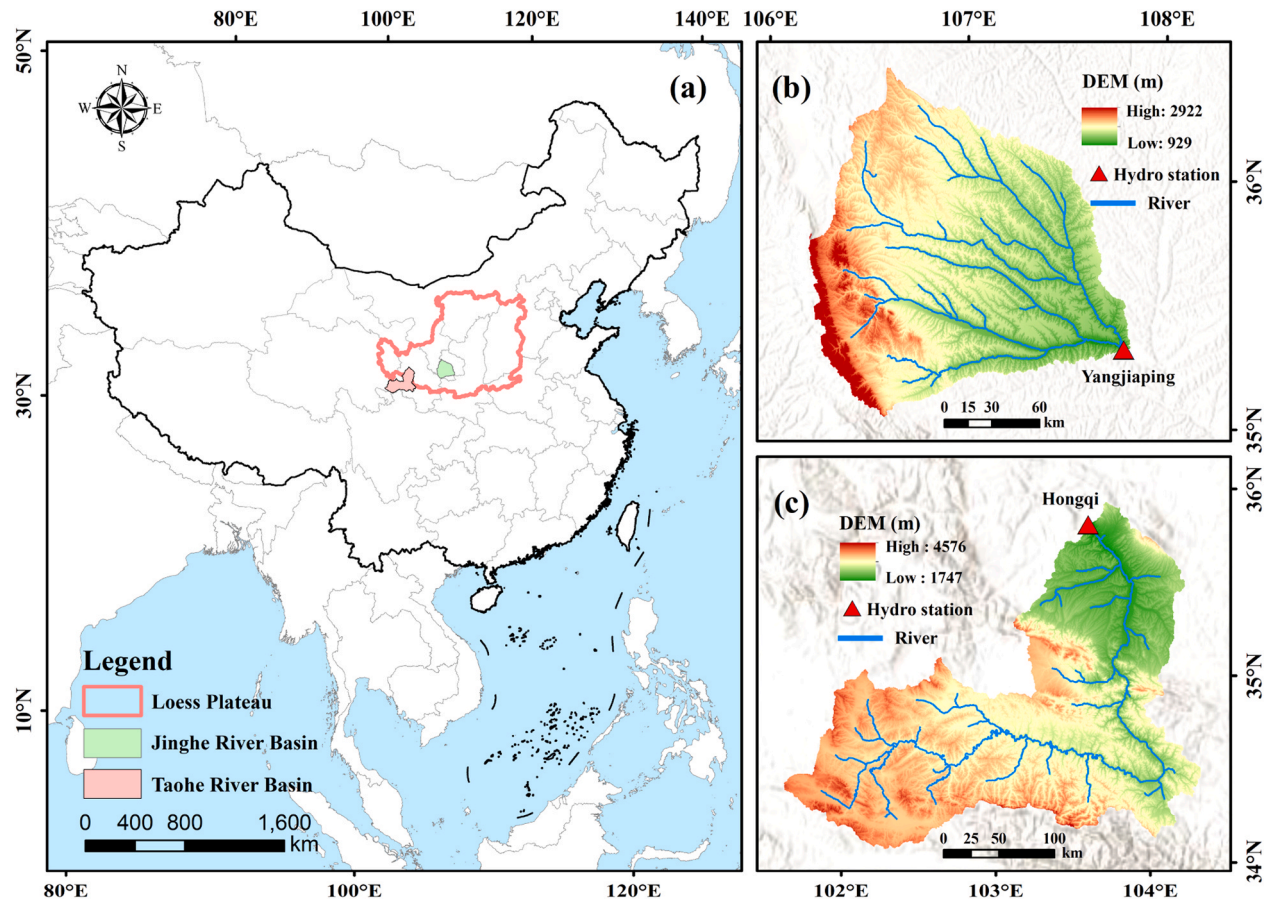


Fig. 1. Study area: (a) location of the Jinghe and Taohe River basins within the Loess Plateau, China; (b–c) digital elevation model (DEM), river network, and hydrological stations in the Jinghe River Basin (Yangjiaping) and the Taohe River Basin (Hongqi), respectively.

$$R = \begin{cases} \frac{(P - \lambda \times S)^2}{P + (1 - \lambda) \times S} & P > \lambda \times S \\ 0 & P \leq \lambda \times S \end{cases} \quad (1)$$

$$S = \frac{25400}{CN} - 254 \quad (2)$$

Where R represents the surface streamflow (mm), P is the observed precipitation (mm), S is the maximum potential soil water retention (mm), CN is the curve number, an index representing the soil moisture retention capacity (dimensionless), and λ is the initial abstraction ratio (dimensionless).

2.3.2. Modified SWAT

To improve the performance of SWAT in the JRB and TRB on the LP, three modifications were introduced based on previous studies and the hydroclimatic characteristics of the study area. These modifications were designed to improve the representation of vegetation dynamics and runoff generation processes under the complex topographic and ecological conditions of the LP.

First, the default vegetation dormancy scheme in SWAT was modified by replacing the day-length threshold method with remotely sensed phenological data, and the simulated leaf area index (LAI) was replaced with MODIS-derived LAI data. The physical meaning of this modification is to improve the representation of seasonal vegetation growth dynamics, which directly influence evapotranspiration, interception, and soil water consumption. Because vegetation growth on the LP is strongly affected by climate variability and ecological restoration, the default SWAT phenology and LAI simulation scheme may not accurately reflect actual vegetation conditions. Therefore, incorporating remotely sensed phenology and LAI data provides a more realistic description of vegetation dynamics and improves the simulation of hydrological processes. Detailed descriptions of this modification can be found in (Jin et al., 2024).

Second, the initial abstraction ratio (λ) was adjusted to account for the effect of slope on rainfall abstraction and runoff generation. In the original SWAT model, λ is fixed at 0.2, which is mainly applicable to relatively flat areas. However, the JRB and TRB are characterized by large terrain fluctuations, steep slopes, and widespread gully landforms, which can substantially affect infiltration and surface runoff generation. Previous studies have shown that λ decreases exponentially with increasing slope (Pang et al., 2020). Therefore, in this study, λ was adjusted using a fitted function derived from regional field observations (Table S1), as expressed in Eq. (3) and illustrated in Fig. S1, so that the model could better represent the topographic control on rainfall partitioning in the study basins.

Third, the empirical curve number (CN) was modified using a slope-adjusted formulation to better describe the runoff response of Hydrological Response Units (HRUs) under complex terrain conditions. The physical meaning of this modification is to account for the influence of slope on runoff generation potential. The default CN values in SWAT were originally developed for relatively flat agricultural watersheds, and may therefore be less suitable for the highly dissected terrain of the Loess Plateau. Because slope strongly affects infiltration opportunity, soil water retention, and overland flow generation, a slope-adjusted CN can provide a more realistic representation of runoff generation in the JRB and TRB. In this study, the slope-adjusted CN formulation proposed by Huang et al. (2006) for the Loess Plateau was adopted to improve streamflow simulation, as shown in Eq. (4).

$$\lambda = 0.294 \times e^{-0.077 \times \text{slope}} \quad (3)$$

$$CN_{\text{slope}} = CN \times \frac{15.63 \times \text{slope} + 322.79}{\text{slope} + 323.52} \quad (4)$$

Where λ represents the initial abstraction ratio, CN and CN_{slope} refer to the original and the slope-adjusted curve numbers, respectively. slope denotes the average slope of the HRU, expressed in degrees (°) in Eq. (3) and in percentage (%) in Eq. (4). Therefore, unit conversion is required during calculation.

2.3.3. Model calibration and validation

Model calibration involves adjusting key parameters to improve agreement between simulated results and observed data and represents an essential component of hydrological modeling (Bawa et al., 2024). In this study, 22 streamflow-related parameters were included in the calibration process (Table S2). Before calibration, a sensitivity analysis was conducted using the SWAT-CUP framework to identify the parameters that exerted the strongest influence on streamflow simulation. The sensitivity ranks, parameter ranges, and optimal values for the original and improved SWAT models in the JRB and TRB are presented in Table S2. Based on the sensitivity analysis, the most influential parameters were prioritized during calibration to reduce parameter uncertainty and improve model robustness.

Multiple rounds of iterative calibration were then performed using SWAT-CUP until the model performance indicators stabilized and showed no further improvement (Liu, 2020; Moriasi, D. et al., 2015). In this way, parameter sensitivity was explicitly considered during calibration, and parameter uncertainty was reduced through sensitivity-guided parameter adjustment.

The Nash–Sutcliffe Efficiency (NSE) and Kling–Gupta Efficiency (KGE) were used as the primary model performance indicators. In addition, Relative Error (RE) and Root Mean Square Error (RMSE) were included as supplementary evaluation metrics to further assess model bias and simulation error. Their calculation methods are as follows:

$$NSE = 1 - \frac{\sum_{i=1}^n (Q_{obs} - Q_{sim})^2}{\sum_{i=1}^n (Q_{obs} - \bar{Q}_{obs})^2} \quad (5)$$

$$KGE = 1 - \sqrt{(CC - 1)^2 + \left(\frac{\bar{Q}_{sim}}{\bar{Q}_{obs}} - 1\right)^2 + \left(\frac{Q_{sim}^{std}}{Q_{obs}^{std}} - 1\right)^2} \quad (6)$$

$$RE = \frac{\bar{Q}_{sim} - \bar{Q}_{obs}}{\bar{Q}_{obs}} \times 100\% \quad (7)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Q_{sim} - Q_{obs})^2} \quad (8)$$

where Q_{obs} and Q_{sim} represent the observed and simulated streamflow, respectively, and $\bar{Q}_{obs}$ and $\bar{Q}_{sim}$ are their corresponding mean values. CC denotes the correlation coefficient of Q_{obs} and Q_{sim} , and i^{std} denotes the standard deviation.

2.3.4. Baseflow separation method

Baseflow was extracted from total streamflow using the Eckhardt digital filtering method (Eq. (9)). Xu et al. (2025) compared seven baseflow separation techniques and found that the Eckhardt approach is particularly well suited to the Yellow River Basin, as it effectively extracts baseflow and captures the hydrological characteristics of the watershed.

$$b_t = \frac{(1 - BFI_{max}) \times a \times b_{t-1} + (1 - a) \times Q_t}{1 - a \times BFI_{max}} \quad (9)$$

Where b_t represents the baseflow at time t (m^3/s), b_{t-1} represents the baseflow at time $t - 1$ (m^3/s), Q_t represents the total streamflow at time t (m^3/s), a is the groundwater recession constant, and BFI_{max} is the maximum baseflow index.

In this study, the recession constant a was determined through recession analysis, with values of 0.868 for the JRB and 0.931 for the TRB. Eckhardt (2008) suggested that BFI_{max} varies with aquifer and streamflow conditions, with typical values of 0.80 for perennial streams with porous aquifers, 0.50 for ephemeral streams with porous aquifers, and 0.25 for perennial streams with hard rock aquifers. Therefore, considering the hydrogeological background of the Loess Plateau, the seasonal hydrological characteristics of the study basins, and previous studies in the Yellow River Basin (Xu et al. (2025)), BFI_{max} was set to 0.5 in this study.

2.3.5. Identification of drought events and characteristics

In this study, meteorological drought was quantified using the Standardized Precipitation Index (SPI), while streamflow and baseflow droughts were assessed based on the corresponding standardized indices (SSI and SBI). These three drought indices follow the same calculation process (Lee and Ajami, 2023; Wang et al., 2024; Wu et al., 2018). The drought characteristics, including drought duration, intensity, and frequency, for different drought types were identified based on run theory (Wu et al., 2017). According to drought classification, a drought threshold of -0.5 was set, and drought events were identified when the drought indices fell below this threshold.

2.3.6. Propagation of drought

The correlation coefficient between SPI and SSI is widely used to calculate the time of drought propagation (Ding et al., 2021a; Shi et al., 2022). To explore how drought transitions from the meteorological phase to the hydrological phase, this study examined the relationship between SPI, SSI, and SBI. Specifically, we selected the 1-month Standardized Streamflow Drought Index (SSI-1) and the Standardized Baseflow Drought Index (SBI-1) because they are more effective in representing short-term hydrological droughts. By analyzing the correlation between SPI at different time scales (SPI- n , where n represents the number of months) and SSI-1 as well as SBI-1, this study further determined the optimal time for drought propagation (in months) and the optimal response intensity (represented by R). The correlation was quantified using the Pearson correlation coefficient (Barker et al., 2016).

2.3.7. Drought event matching

In fact, meteorological drought events do not always evolve into hydrological droughts. To identify which meteorological droughts may have triggered specific hydrological droughts, a matching analysis between the two types of events is necessary. This study adopts the methodology proposed by (Guo et al., 2020) and uses Eq. (10) to determine the triggering time lag between drought events.

$$trigger\ interval = \begin{cases} (ts_i - T_p, te_i), & \text{if } ts_i - ts_{i-1} \geq T_p \\ (te_{i-1}, te_i), & \text{if } ts_i - ts_{i-1} < T_p \end{cases} \quad (10)$$

Where T_p denotes the propagation time derived from the correlation analysis in Section 2.3.6. For a given sub-basin and drought propagation type, T_p is treated as a predefined parameter rather than an event-specific dynamic variable. Eq. (8) is used to define the triggering window of a hydrological drought event for matching it with a preceding meteorological drought event. A meteorological drought is deemed to induce a corresponding hydrological drought when its onset falls within the hydrological drought's triggering window.

The effective propagation rate is calculated as the proportion of meteorological drought events that trigger hydrological droughts to all meteorological drought events within a given period (Eq. (11)).

$$PR = \frac{MD_e}{MD_t} \quad (11)$$

Where PR denotes the drought propagation rate, MD_e is the count of meteorological drought events leading to hydrological droughts, and MD_t denotes the total number of meteorological drought events.

3. Results

3.1. Model validation and performance enhancement for SWAT

The optimal parameter values derived from model calibration were then used to simulate streamflow. Streamflow hydrographs simulated by the original and improved SWAT models are shown in Fig. 2. The results demonstrate that the improved SWAT enhanced simulation performance during both the calibration and validation periods in the two basins. In the JRB, during the calibration period, NSE increased from 0.873 to 0.886 and KGE increased from 0.880 to 0.906, while RE decreased in magnitude from -7.762% to -1.987% and RMSE decreased from 5.030 to 4.748 (Table 1). During the validation period, NSE increased from 0.758 to 0.882 and KGE increased from 0.693 to 0.845, while RMSE decreased from 7.316 to 5.112, indicating a clear improvement in model performance, although the absolute RE increased slightly from 2.056% to 3.890% (Table 1). In addition, comparison of the streamflow hydrographs showed that the original SWAT significantly overestimated the streamflow peak in 2018, whereas the improved SWAT captured this peak more accurately (Fig. 2a), suggesting greater model stability after calibration.

In the TRB, the improved SWAT also showed better performance. During the calibration period, NSE increased from 0.767 to 0.788 and KGE increased from 0.805 to 0.829, while the absolute RE decreased from 1.826% to 1.612% and RMSE decreased from 43.624 to 41.636 (Table 1). During the validation period, NSE increased from 0.789 to 0.804 and KGE increased from 0.848 to 0.868, and RMSE decreased from 46.238 to 44.618, and RE decreased from 8.096% to 3.211% (Table 1). Moreover, the hydrograph simulated by the improved model showed better agreement with the observed streamflow, reflecting a more realistic representation of hydrological processes (Fig. 2b). Overall, the improved SWAT produced greater improvement in streamflow simulation performance in the JRB than in the TRB.

To further evaluate the performance of the original and improved SWAT models in simulating both streamflow and baseflow

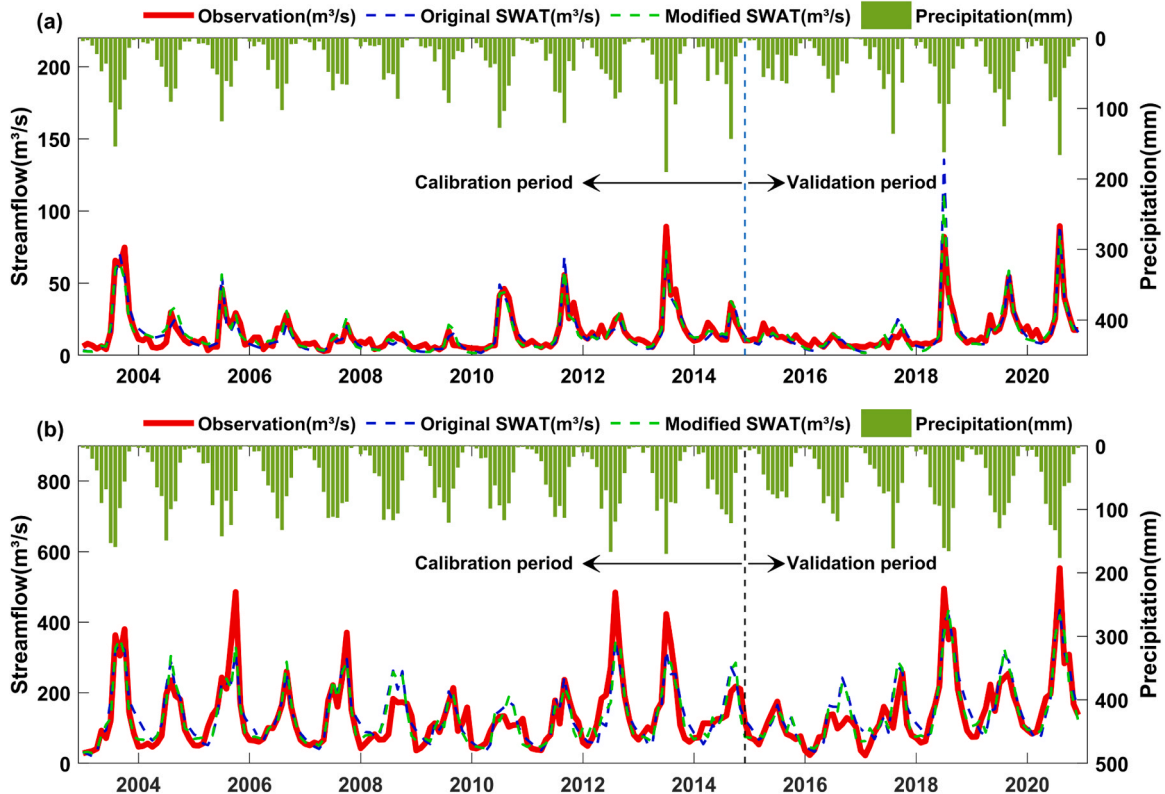


Fig. 2. Process curves of streamflow simulated by the improved and original SWAT models compared with observed streamflow. (a): JRB; (b): TRB.

Table 1

Performance of SWAT and the improved SWAT in streamflow simulations.

Watershed	Model	calibration period (2003–2014)				validation period (2015–2020)			
		NSE	KGE	RE	RMSE	NSE	KGE	RE	RMSE
JRB	SWAT	0.873	0.880	-7.762	5.030	0.758	0.693	-2.056	7.316
	Modified SWAT	0.886	0.906	-1.987	4.748	0.882	0.845	-3.890	5.112
TRB	SWAT	0.767	0.805	1.826	43.624	0.789	0.848	8.096	46.238
	Modified SWAT	0.788	0.829	-1.612	41.636	0.804	0.868	3.211	44.618

(Table S3), scatter plots were constructed by comparing observed and simulated streamflow as well as Eckhardt-separated observed and simulated baseflow for the JRB and TRB (Fig. 3). For streamflow, the improved SWAT showed better agreement with observations in both basins, with NSE increasing from 0.832 to 0.885 and KGE increasing from 0.877 to 0.937 in the JRB (Fig. 3a), and NSE increasing from 0.777 to 0.795 and KGE increasing from 0.831 to 0.850 in the TRB (Fig. 3b). The scatter points of the improved SWAT were generally closer to the fitted line, indicating that the model modifications improved the simulation of streamflow dynamics, particularly in the JRB.

A similar comparison was conducted for baseflow using the Eckhardt-separated series. In both basins, the simulated baseflow values showed positive correlations with the observed baseflow, suggesting that the models were able to reproduce the general variability of baseflow. In the JRB, the improved SWAT produced a significantly increase in KGE from 0.900 to 0.924, while NSE also increased slightly from 0.849 to 0.854 (Fig. 3c). In the TRB, the improved SWAT showed a more evident improvement in baseflow

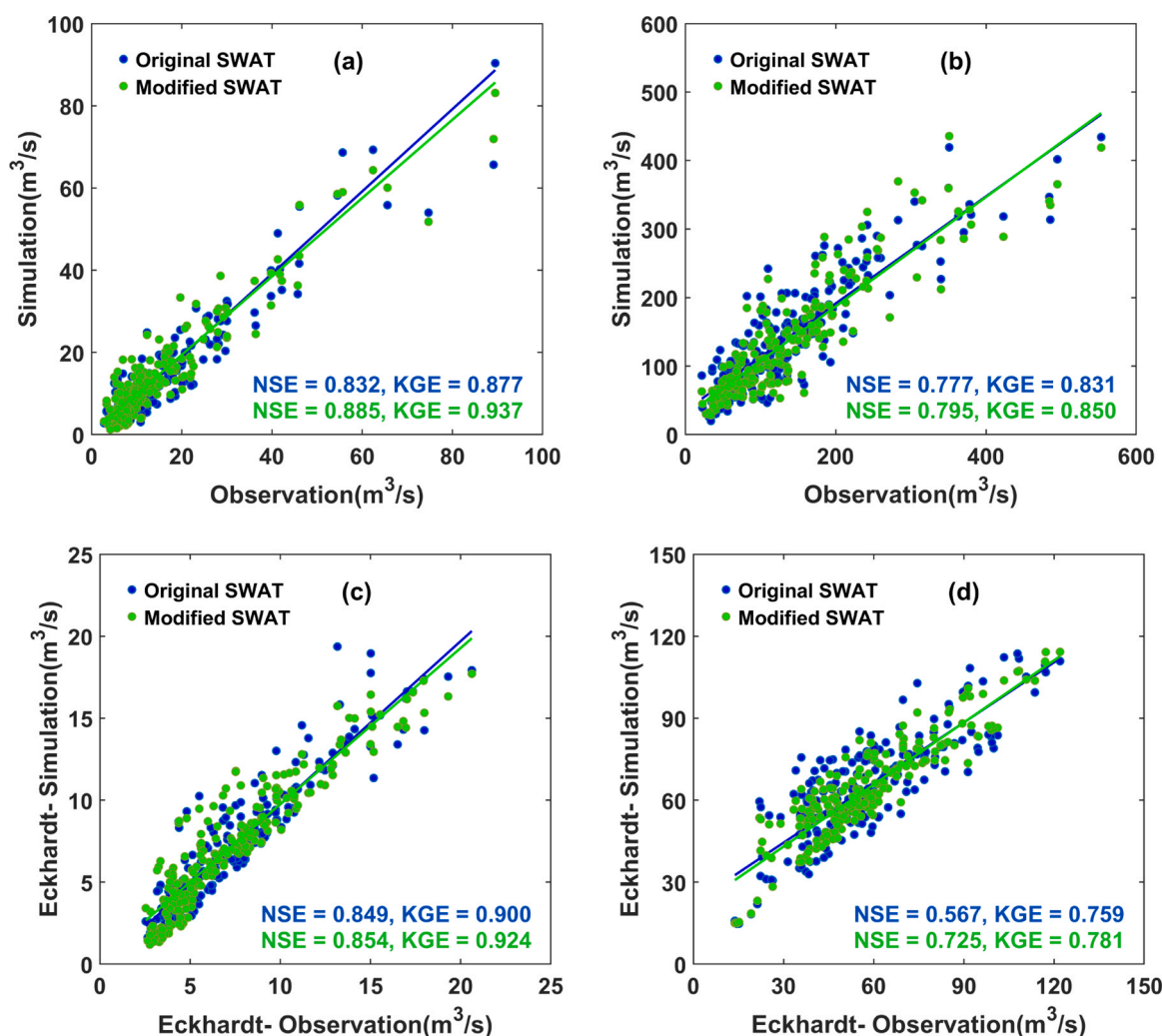


Fig. 3. Scatter plots comparing observed and simulated streamflow and baseflow for the original and improved SWAT models in the JRB and TRB. (a) Streamflow in the JRB; (b) Streamflow in the TRB; (c) Baseflow in the JRB; (d) Baseflow in the TRB.

simulation, with NSE increasing from 0.567 to 0.725 and KGE increasing from 0.759 to 0.781 (Fig. 3d). Overall, the improved SWAT enhanced streamflow simulation in both basins and provided a reasonable basis for baseflow simulation, thereby supporting its subsequent application in hydrological drought and drought propagation analysis.

3.2. Spatial distribution of streamflow and baseflow

To visually illustrate the spatial distribution characteristics of streamflow and baseflow in the JRB and TRB, this study conducted a spatial visualization analysis of the multi-year annual average streamflow and baseflow simulated by the improved SWAT (Fig. 4). The spatial analysis results indicate that the annual average streamflow in the JRB increases gradually from west to east and from north to south. The annual average streamflow in low-flow areas primarily ranges between 0 and 2 m^3/s , while sub-basins near the total outlet in the downstream area of the basin have an annual average streamflow exceeding 4 m^3/s (Fig. 4a). The spatial distribution characteristics of baseflow are similar to those of streamflow, both exhibiting a significant elevation dependence. The annual average baseflow in low-flow areas primarily ranges between 0 and 0.6 m^3/s , while sub-basins near the total outlet in the downstream area of the basin have an annual average baseflow greater than 2 m^3/s (Fig. 4b). The spatial distribution of annual average streamflow and baseflow in the TRB shows similar trends, both exhibiting a gradual increase from west to east and from south to north (Fig. 4c and d).

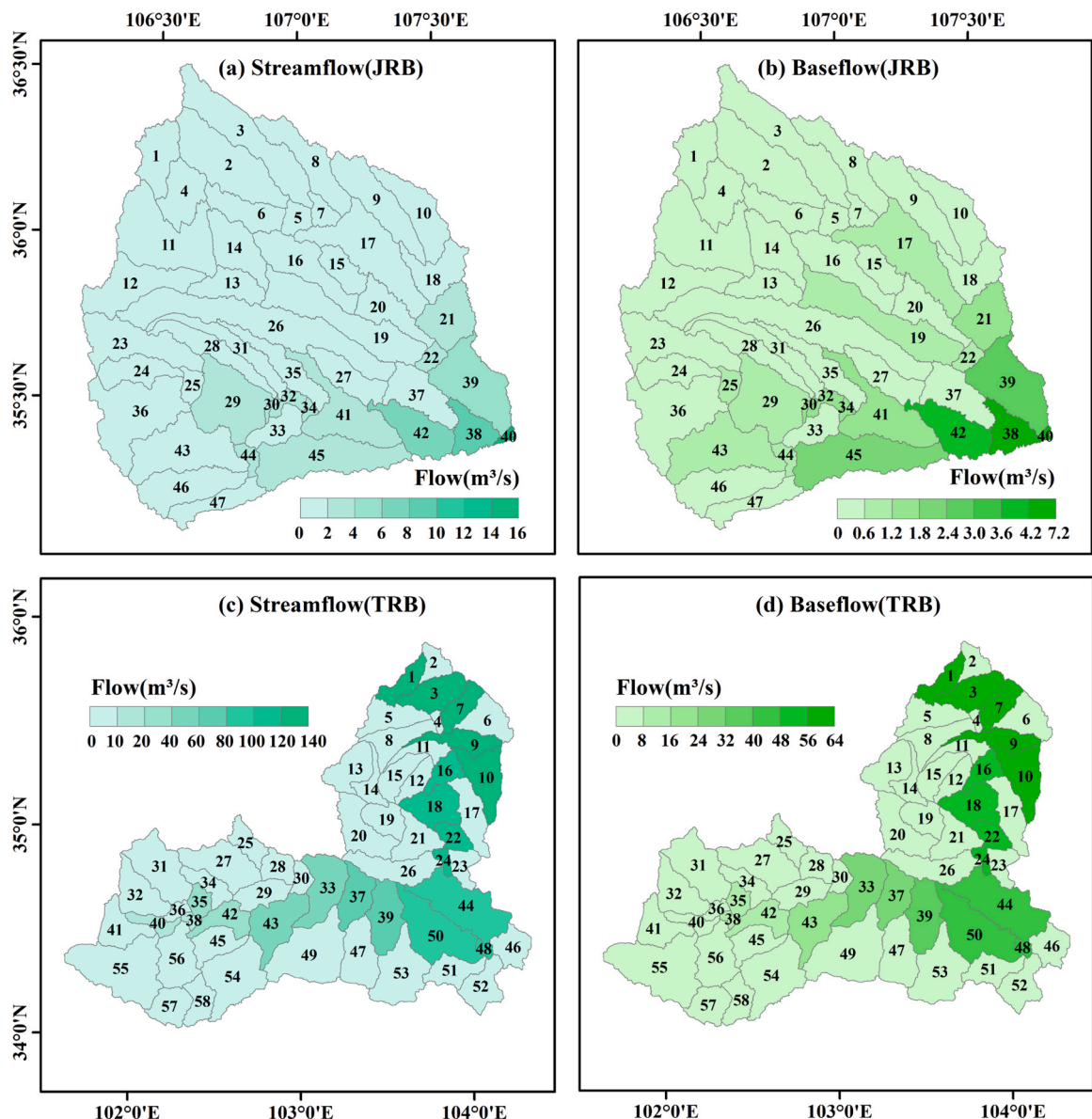


Fig. 4. Spatial distribution of streamflow and baseflow simulated by the improved SWAT.

The annual average streamflow in low-flow areas primarily ranges between 0 and 20 m³/s, while the annual average baseflow is mainly between 0 and 16 m³/s. In the high-flow areas of the middle and lower reaches of the basin, the annual average streamflow exceeds 100 m³/s, and the annual average baseflow exceeds 48 m³/s. Overall, the streamflow and baseflow in both basins exhibit a spatial distribution pattern that gradually accumulates from the mountains to the plains and from the upstream to the downstream, reflecting the regional differences in the hydrological characteristics of the basins and the spatial heterogeneity of their water resources.

3.3. Characteristics of meteorological and hydrological droughts

3.3.1. Spatial patterns of meteorological drought

The spatial distribution characteristics of meteorological drought in each sub-basin were obtained through the SPI calculated from precipitation data, combined with the analysis of drought frequency, duration, and intensity using run theory (Fig. 5). In the JRB, the frequency of meteorological drought occurrences is relatively high, mainly ranging between 28 and 32 times, with the most frequent drought events occurring in the central and western regions of the basin (Fig. 5a). The duration of drought events is generally short, concentrated between 1.6 and 2.0 months, and its spatial distribution follows a trend similar to that of drought frequency (Fig. 5b). The majority of drought intensities are concentrated between 1.3 and 1.5, indicating that although drought events are frequent in the basin, the intensity of each drought is relatively low, with a relatively mild impact (Fig. 5c). Overall, the JRB is characterized by frequent droughts, short durations, and moderate intensities.

In contrast, the frequency of meteorological drought occurrences in the TRB is relatively low, primarily ranging between 22 and 28 times, with a distinct spatial variation. The drought frequency is lower in the upstream areas, while higher frequencies are observed in the middle and lower reaches of the basin (Fig. 5d). The drought duration in the TRB is relatively long, primarily ranging between 2.2 and 2.8 months (Fig. 5e). In terms of drought intensity, the TRB generally experiences higher intensity, especially in the upstream areas, where the drought intensity is concentrated between 1.5 and 1.7, indicating that the region has experienced more severe drought events. In contrast, the drought intensity in the middle and lower reaches is lower, ranging between 1.1 and 1.5, suggesting that these areas have experienced milder drought impacts, which may be related to local precipitation and water resource availability (Fig. 5f).

3.3.2. Spatial patterns of streamflow and baseflow drought

Based on the streamflow and baseflow data simulated by the improved SWAT, the streamflow drought index and baseflow drought index for the JRB and TRB were calculated. Using run theory, the drought frequency, duration, and intensity for both streamflow and baseflow droughts were further extracted. The spatial distribution of streamflow drought characteristics is shown in Fig. 6. In the JRB, the frequency of streamflow drought occurrences is mainly concentrated between 9 and 15 times, with a higher frequency observed in the western part of the basin (Fig. 6a). The duration of streamflow droughts is generally long, with droughts lasting over 5 months in most areas of the basin, and the spatial distribution of drought duration closely resembles that of drought frequency (Fig. 6b). The intensity of streamflow droughts is relatively high, with drought intensity exceeding 3.2 in most areas of the basin, and in some regions, drought intensity even exceeds 4.8 (Fig. 6c).

Compared to the JRB, the occurrence frequency of streamflow drought in the TRB is generally higher. In the middle and lower reaches, the drought frequency exceeds 13 times, with some areas experiencing drought frequencies ranging from 19 to 21 times (Fig. 6d). However, the duration of streamflow droughts in the TRB is relatively short, with the drought duration across the entire basin mainly concentrated between 3 and 6 months (Fig. 6e). Additionally, the intensity of streamflow droughts in the TRB is relatively low, primarily ranging from 1.6 to 4.0 (Fig. 6f).

The spatial distribution of baseflow drought characteristics is shown in Fig. 7. In the JRB, the frequency of baseflow droughts generally ranges from 5 to 7 times. Only in certain areas of the northern and western parts of the basin is the frequency of droughts relatively high, reaching 7–9 times. In addition, there are several scattered sub-basins within the basin, mainly in the central-southern part, where the frequency of baseflow droughts is less than 5 times (Fig. 7a). The duration of baseflow droughts is generally long, with values predominantly ranging from 10 to 16 months across the basin. Relatively longer drought durations are mainly observed in the southern part of the basin, whereas shorter durations are found in parts of the northern and central areas (Fig. 7b). The spatial pattern of baseflow drought intensity largely mirrors that of drought duration. In most areas, intensity values range from 5.1 to 7.7, while relatively higher intensities are mainly observed in the southern and southeastern parts of the basin, and lower intensities are found in some northern and central sub-basins (Fig. 7c).

Compared to the JRB, the occurrence frequency of baseflow droughts in the TRB shows significant spatial variation. The drought frequency is higher in the southern and southeastern parts of the basin, concentrated between 7 and 11 times, while the other parts of the basin experience lower drought frequencies, mainly ranging from 3 to 7 times (Fig. 7d). The duration of baseflow droughts in the TRB is relatively short, particularly in the western and central parts, where the drought duration is typically between 3 and 10 months. In contrast, the drought duration in the eastern and northeastern parts is longer, usually ranging from 10 to 19 months (Fig. 7e). The spatial distribution of baseflow drought intensity closely aligns with that of drought duration. The drought intensity is relatively low in the western and central parts of the basin, generally ranging from 2.5 to 5.1, whereas higher intensities are observed in the eastern and northeastern parts, where values typically exceed 5.1 (Fig. 7f).

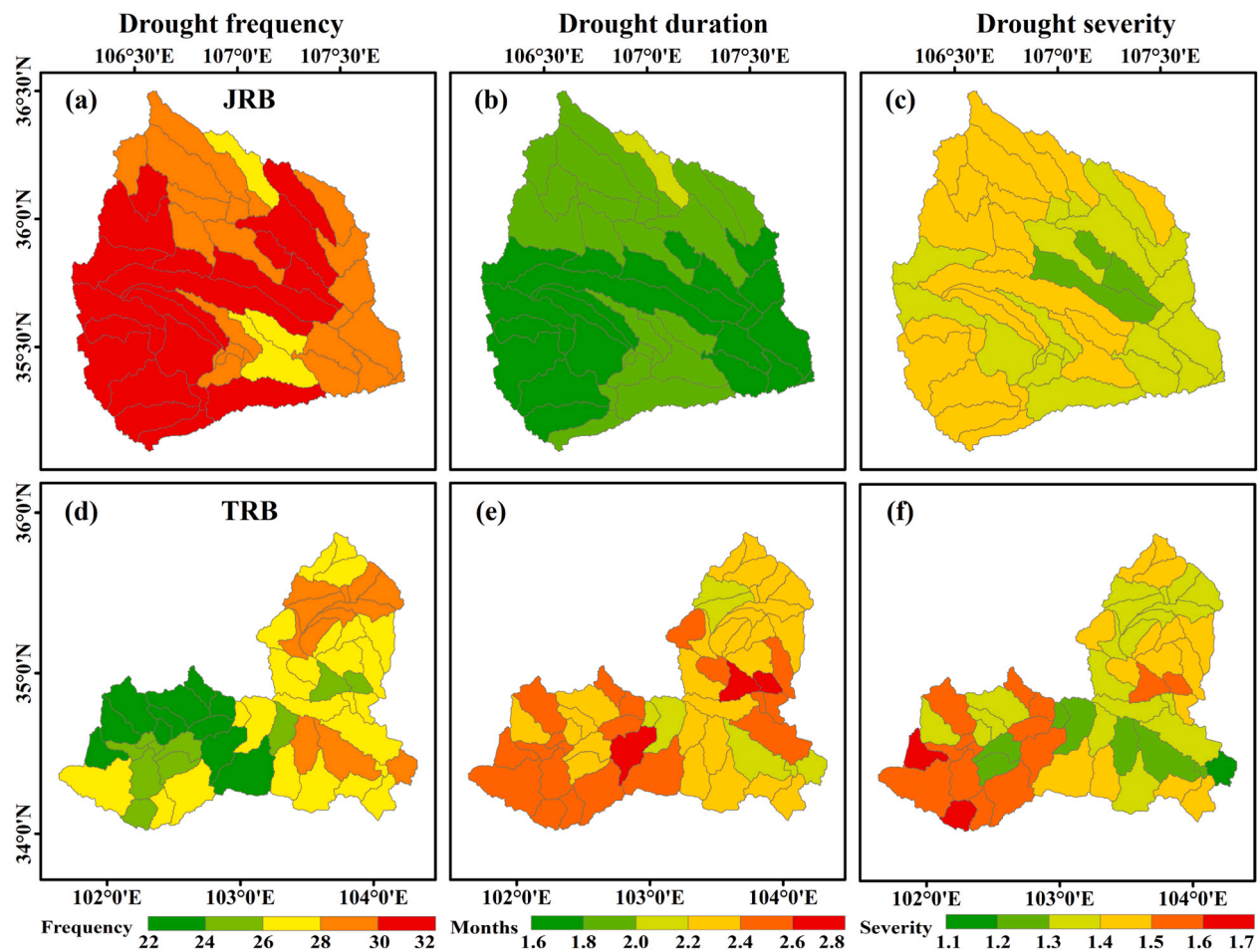


Fig. 5. Spatial distribution of meteorological drought characteristics.

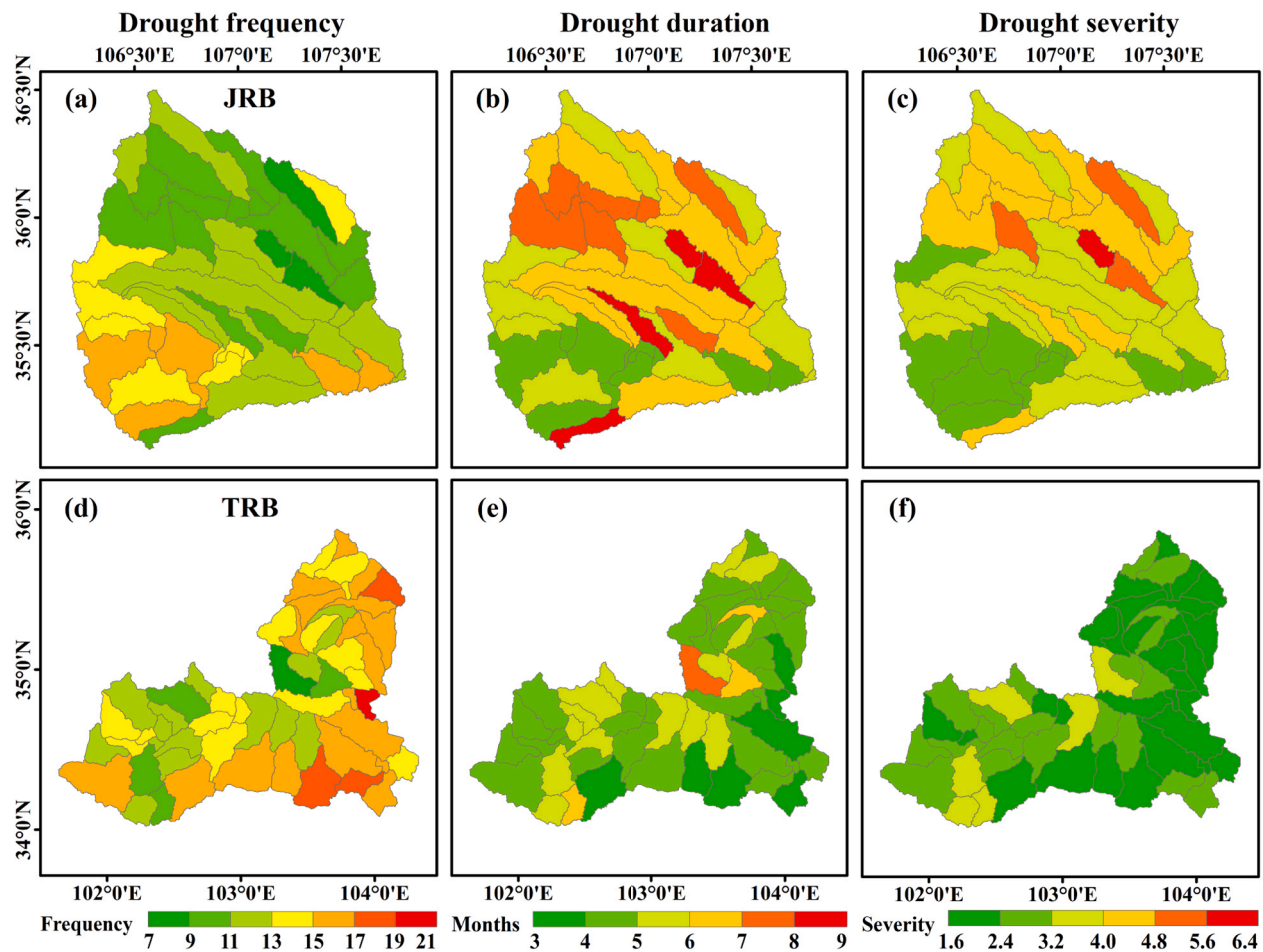


Fig. 6. Spatial distribution of streamflow drought characteristics.

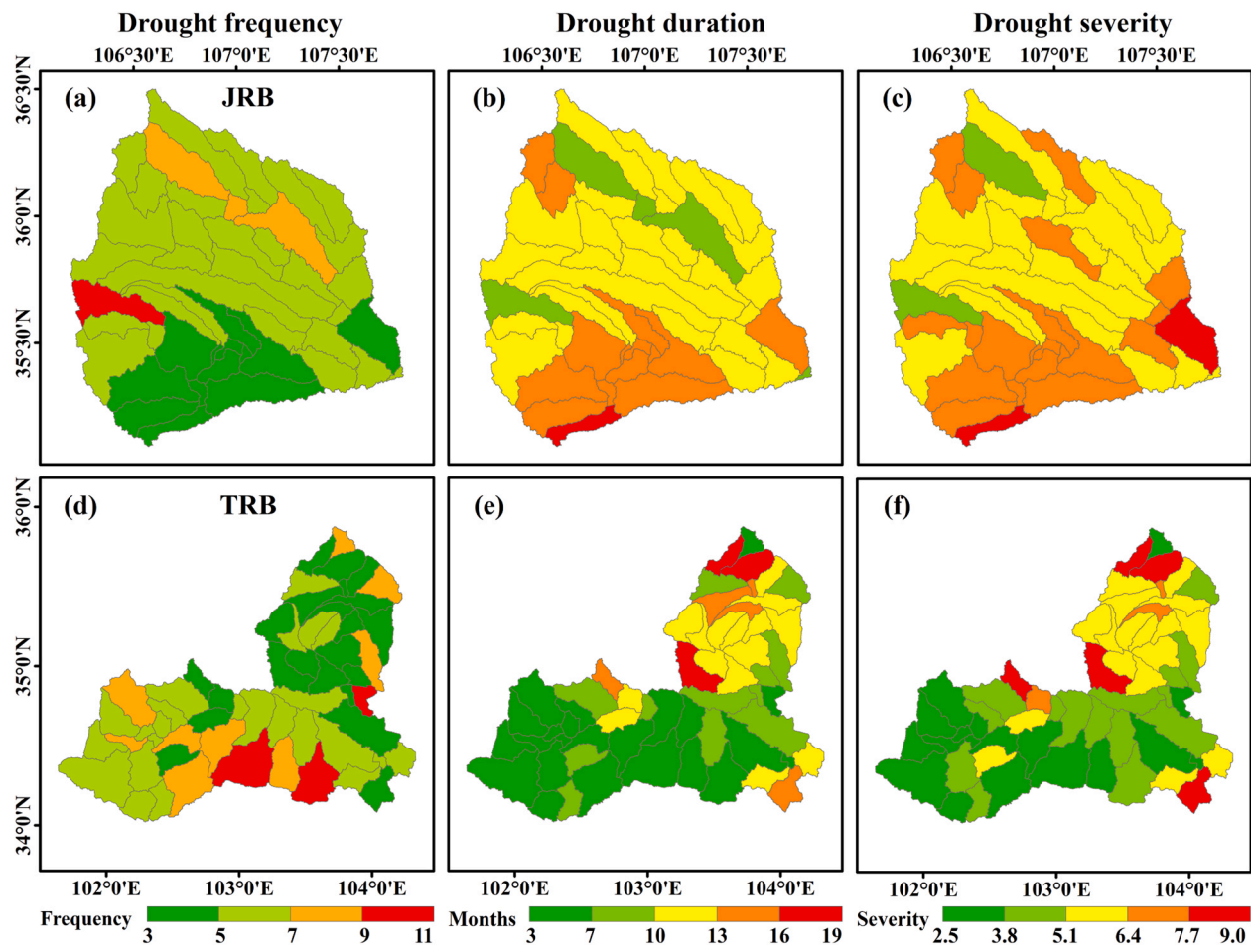


Fig. 7. Spatial distribution of baseflow drought characteristics.

3.4. Drought propagation in sub-basins

3.4.1. Propagation timing from meteorological to hydrological drought

This study, based on meteorological, streamflow, and baseflow drought indices, investigates the propagation time (PT) of meteorological drought to hydrological drought through correlation analysis. Fig. 8a shows the spatial distribution of PT from meteorological drought to streamflow drought in the JRB. The PT is mainly concentrated between 7 and 12 months. Compared with other areas of the basin, PT is relatively longer in some northwestern sub-basins. This suggests that the hydrological response in the northwest of the basin is slower, potentially influenced by the region's complex topography and hydrological conditions, resulting in a longer conversion time from meteorological drought. In contrast, the PT of meteorological drought to baseflow drought is mainly concentrated between 12 and 14 months, showing a trend of earlier propagation from the central part towards the north and south (Fig. 8b). Overall, the spatial distribution of the PT of meteorological drought to baseflow drought differs significantly from that to streamflow drought, with the PT to baseflow drought generally being longer. Fig. 9a shows the spatial distribution of the difference in PT from meteorological drought to streamflow drought and baseflow drought in the JRB. The results indicate that the PT difference is generally small in the northwest of the basin, with a trend of gradually increasing towards the southeast. However, some sub-basins in the northwest (sub-basins 2 and 3) exhibit a larger difference in PT, suggesting that the drought propagation characteristics in localized

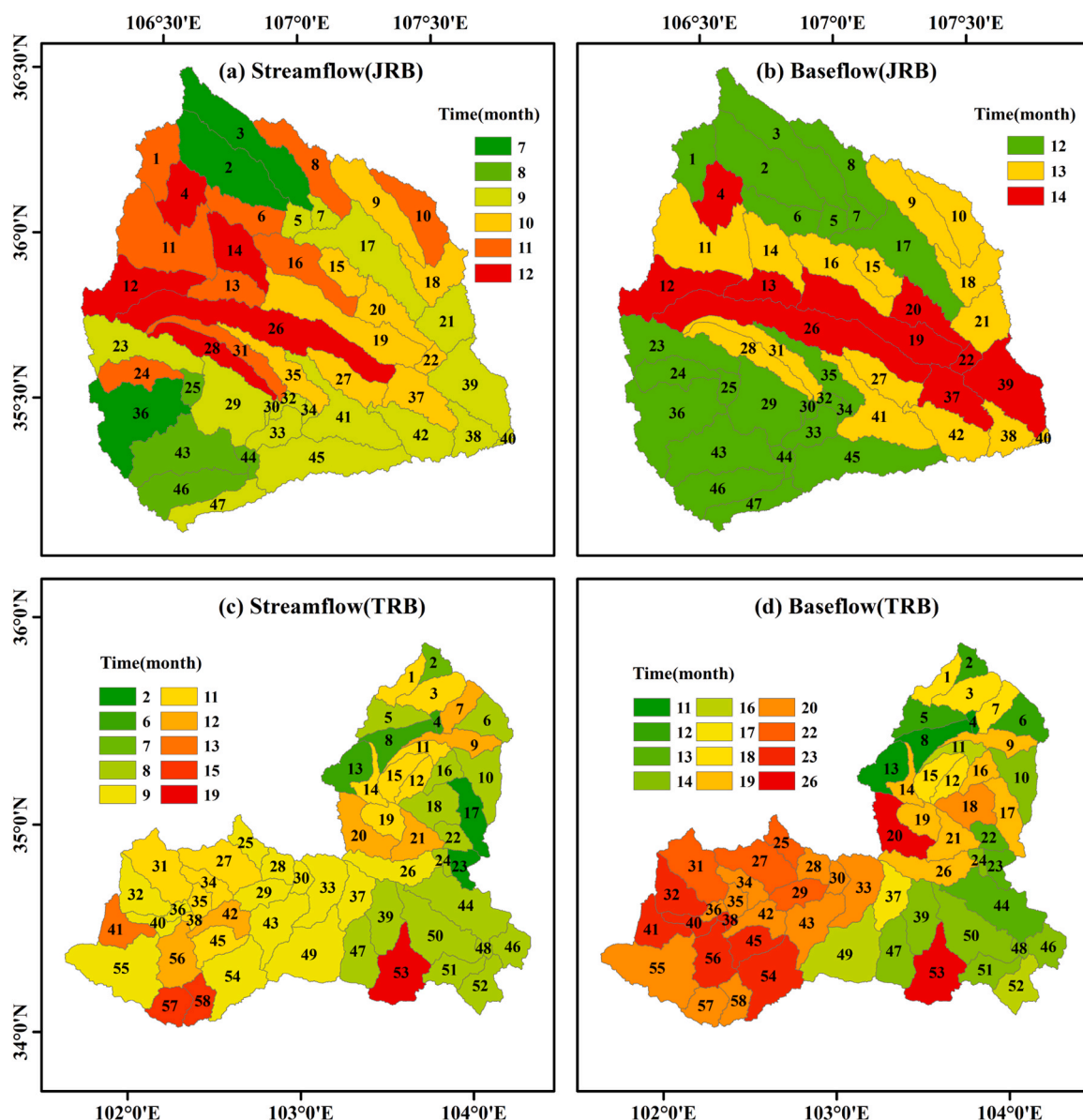


Fig. 8. Spatial distribution of the propagation time of meteorological drought to streamflow and baseflow droughts.

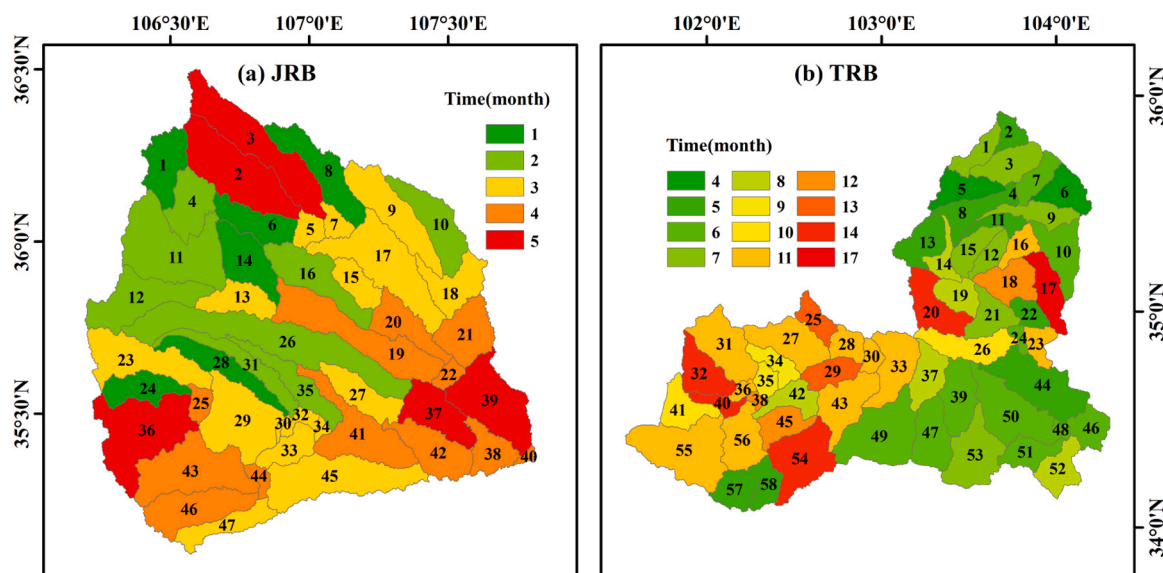


Fig. 9. Time difference between meteorological drought propagation to baseflow and streamflow droughts.

areas may be influenced by additional factors.

Fig. 8c shows the spatial distribution of the PT from meteorological drought to streamflow drought in the TRB. Compared to the JRB, the PT in the TRB exhibits more significant spatial distribution differences, ranging from 2 months to 19 months. The majority of the basin's regions have drought propagation times concentrated between 7 and 9 months, while a few sub-basins show considerable deviations. For example, the PT in Sub-basins 17 and 23 is only 2 months, while in Sub-basins 53, 57, and 58, the PT extends up to 19 months. Fig. 8d illustrates the spatial distribution of the PT from meteorological drought to baseflow drought in the TRB. The range of PT is from 11 to 26 months, with the western upstream areas of the basin exhibiting longer PT, primarily concentrated between 20 and 26 months. The PT in the central part of the basin is mainly between 11 and 14 months, while most of the downstream areas show PT ranging from 16 to 19 months. Fig. 9b shows the spatial distribution of the difference in PT between meteorological drought and streamflow drought, as well as baseflow drought, in the TRB. The PT differences in the middle and lower reaches of the basin are mainly concentrated between 4 and 7 months, while the upstream areas exhibit larger differences, concentrated between 10 and 15 months. Although there is some correlation between the PT difference and elevation, this relationship is not as pronounced as in the JRB. This suggests that the drought propagation process in the TRB may be influenced by multiple factors, including not only elevation but also hydrological conditions, vegetation cover, and other factors within the basin.

Overall, there are significant spatial differences in PT between the JRB and TRB. In the JRB, the PT is relatively uniform, with longer PT in some sub-basins in the northwest. In contrast, the TRB exhibits greater regional variability, particularly in the PT from meteorological drought to baseflow drought. These results indicate that topography, precipitation patterns, and hydrological characteristics have a significant impact on drought propagation.

3.4.2. Intensity of streamflow and baseflow droughts responses to meteorological drought

Fig. 10 shows the response intensity of streamflow drought and baseflow drought to meteorological drought in the JRB and the TRB. In the JRB, the response intensity of streamflow drought to meteorological drought is lower compared to that of baseflow drought (Fig. 10a and b). This is because streamflow changes are mainly directly influenced by variations in precipitation. During meteorological droughts, reduced precipitation leads to a decline in streamflow, and when meteorological conditions recover, streamflow also increases rapidly. Therefore, the response of streamflow is characterized by being short-term and direct, with a weaker response intensity. In contrast, the response of baseflow drought to meteorological drought is more significant, with the response intensity mainly concentrated above 0.7. Changes in baseflow reflect the gradual depletion of groundwater, and its response to reduced precipitation is relatively slow, often requiring some time to become evident. Therefore, when baseflow drought occurs, severe water shortages have often already developed.

The streamflow and baseflow response patterns in the TRB are similar to those in the JRB. As shown in Fig. 10c, the response of streamflow drought to meteorological drought in the TRB is also relatively rapid with a lower response intensity. Fig. 10d shows that the response intensity of baseflow drought to meteorological drought in the TRB is higher, with the response intensity primarily concentrated above 0.85. However, the 8th and 13th sub-basins in the downstream areas of the watershed exhibit significant differences, with response intensities ranging from 0.4 to 0.55. A comparison of the results between the JRB and the TRB reveals similar characteristics in their drought response intensities. Specifically, higher altitudes exhibit more significant responses to meteorological droughts. Additionally, the streamflow response in the JRB is more pronounced, which may be related to the region's precipitation instability and topographical features. However, in both the JRB and the TRB, the response intensity of baseflow is relatively high. Its

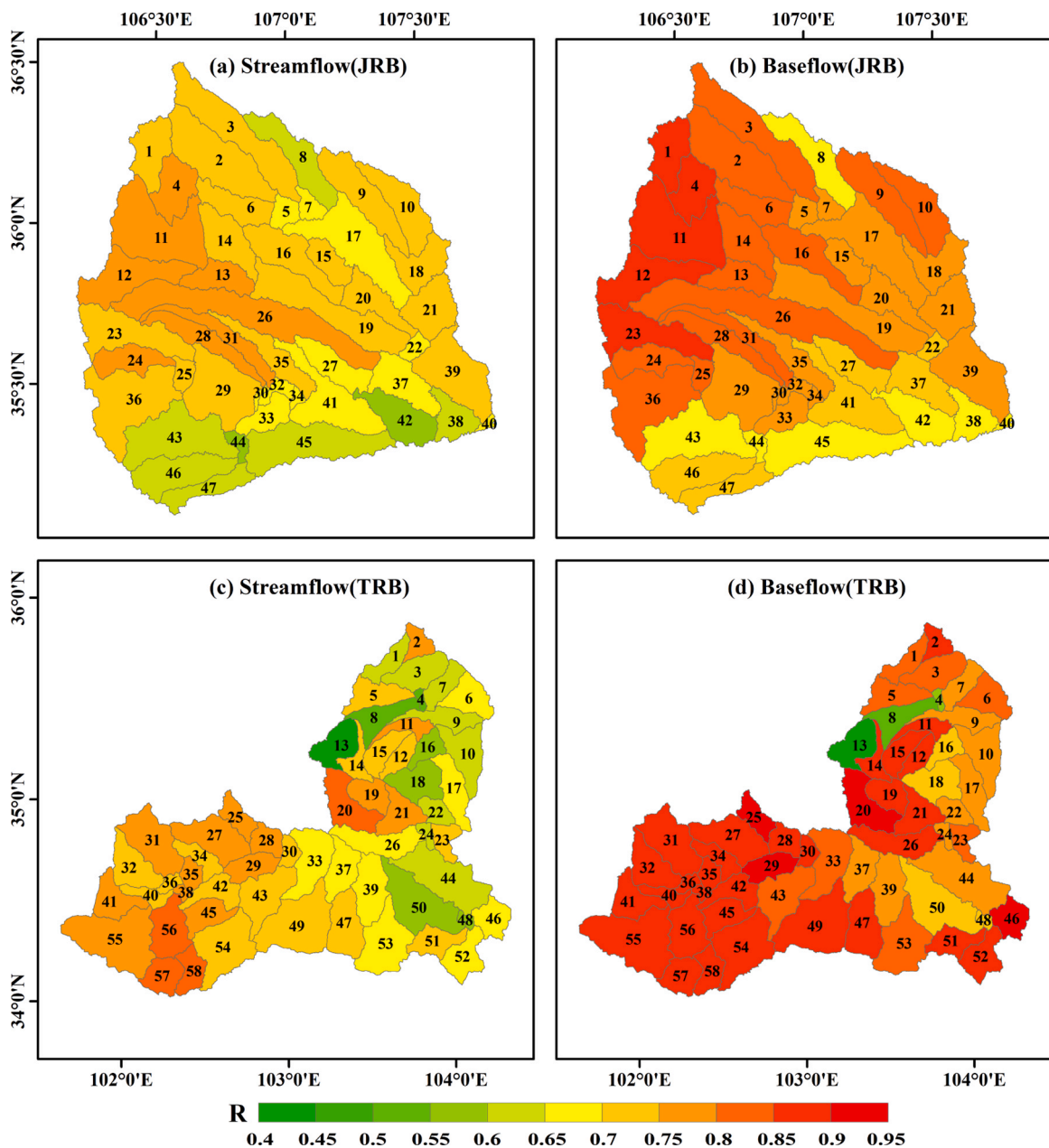


Fig. 10. Spatial distribution of the response intensity of streamflow and baseflow droughts to meteorological drought.

variation is closely linked to long-term precipitation deficits, further confirming the important role of baseflow in meteorological droughts.

3.5. The effective propagation rate of meteorological to hydrological drought

This study matches meteorological and hydrological drought events using the calculated drought propagation time and analyzes the drought propagation rate. The results show that the drought propagation rates in both the JRB and the TRB exhibit significant spatial variability (Fig. 11). In both basins, the effective propagation rate of meteorological drought to streamflow drought (PR-SF) is generally higher than that of meteorological drought to baseflow drought (PR-BF), indicating that streamflow drought is more sensitive to the response of meteorological drought. In the JRB, the PR-SF is primarily concentrated above 65%, with only a few sub-basins (including parts of the middle reaches and sub-basins 2 and 3 in the northern part of the basin) exhibiting lower propagation rates, ranging from 50% to 60% (Fig. 11a). Additionally, the PR-SF in sub-basin 40, which serves as the main outlet of the basin, is also at a relatively high level. The spatial distribution pattern in the TRB is similar to that in the JRB, with PR-SF exceeding 65% in most areas

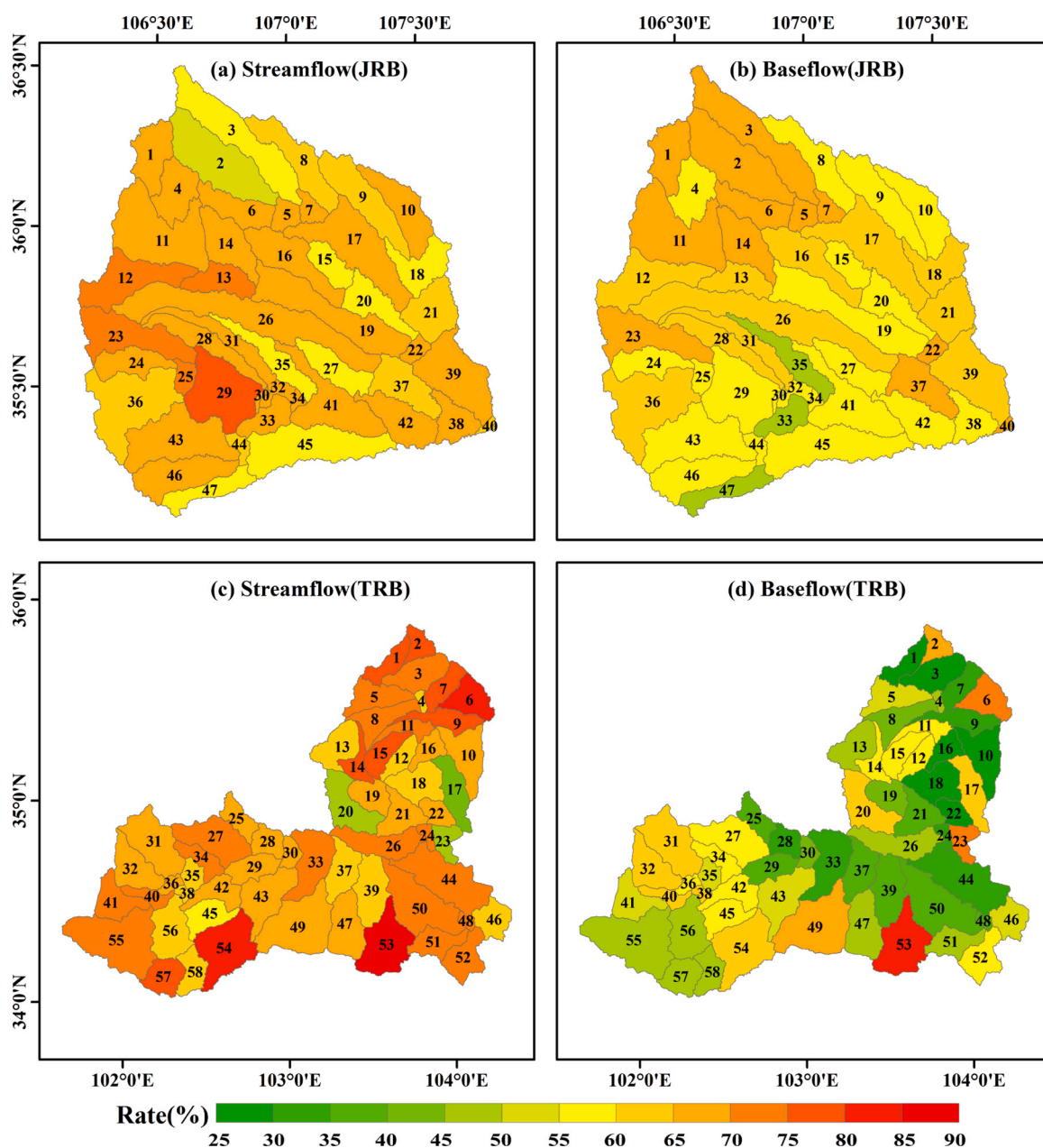


Fig. 11. Spatial distribution of the effective propagation rate of meteorological drought to streamflow and baseflow droughts.

(Fig. 11c). Only a few sub-basins exhibit PR-SF lower than 60%, and these areas are sparsely distributed throughout the entire basin, without showing a clear concentration pattern.

In contrast, the PR-BF exhibits similar spatial distribution characteristics in both basins, with drought propagation rates in the upstream areas significantly higher than those in the middle and lower reaches (Fig. 11b and d). This suggests that baseflow in the upstream areas responds more directly to meteorological drought, while the middle and lower reaches may be influenced by factors such as groundwater recharge, topography, and land use, leading to a reduced propagation rate of baseflow to meteorological drought. This characteristic is particularly prominent in the TRB, where the difference in PR-BF between the upstream and downstream areas is large, indicating a stronger non-uniformity in the propagation of drought within the hydrological system of this basin. Additionally, in the TRB, sub-basin 1, as the main outlet of the basin, shows a significant reduction in PR-BF compared to the upstream areas, which may be influenced by the basin's confluence processes and water resource management measures. Overall, the study indicates that not all meteorological droughts trigger hydrological droughts, and there are significant differences in the response of various hydrological variables to meteorological drought. Streamflow responds more directly to meteorological drought, while baseflow is regulated by more complex hydrological processes, leading to more pronounced spatial heterogeneity in the propagation rate.

4. Discussion

4.1. Differences in drought characteristics and propagation

Compared with previous SWAT-based drought studies, the present study places greater emphasis on improving the reliability of hydrological simulation before conducting drought propagation analysis. Previous studies have used SWAT-simulated streamflow to investigate hydrological drought characteristics and the propagation from meteorological drought to hydrological drought (Chen et al., 2025; Du et al., 2025; Zou et al., 2024). Recent reviews have further pointed out that hydrological models are important tools for drought propagation research, although their results depend strongly on simulation accuracy and remain subject to uncertainty (Zhou et al., 2024). In this context, the improved SWAT adopted in this study provides a more reliable basis for drought characterization and propagation analysis by improving streamflow simulation performance in the two basins. Moreover, unlike most previous studies that mainly focused on streamflow drought, this study extends the analysis to baseflow drought, particularly the propagation from meteorological drought to baseflow drought.

We compared the characteristics and propagation patterns of meteorological drought, streamflow drought, and baseflow drought in the JRB and TRB. The results show that meteorological droughts occur more frequently than hydrological droughts, whereas hydrological droughts generally exhibit longer duration and greater severity (Fig. 12a-c). This result is in agreement with the findings of Huang et al. (2025) in their study of meteorological and hydrological droughts in the Yellow River basin. Meanwhile, our study also reveals the differences between various types of droughts, especially the behavior of baseflow droughts. Existing research has mainly focused on streamflow drought analysis, with baseflow drought often being overlooked. We found that the frequency of baseflow droughts is lower compared to meteorological and streamflow droughts, with the frequency of baseflow drought occurrences in both basins not exceeding 10 times. However, each instance of baseflow drought lasts longer and exhibits greater intensity. The study by Lee and Ajami (2023) also indicates that baseflow droughts are more severe compared to meteorological droughts, but they did not directly compare baseflow droughts with streamflow droughts. This study further explores the differences between baseflow and streamflow droughts and highlights the importance of baseflow droughts in hydrological drought analysis.

The results also reveal clear differences in propagation time among drought types. In both the JRB and TRB, the propagation time from meteorological drought to baseflow drought is notably longer than that from meteorological drought to streamflow drought, while the effective propagation rate to baseflow drought is lower (Fig. 12d-f). This difference is mainly attributed to the relatively rapid response of streamflow to precipitation deficits, whereas baseflow is more strongly regulated by subsurface storage, groundwater recharge, and delayed catchment response (Carroll et al., 2025; Ghaneei et al., 2026). In other words, drought signals must first pass-through precipitation deficit, soil moisture depletion, and subsurface storage reduction before affecting groundwater-supported discharge and eventually baseflow, resulting in a longer propagation lag and a weaker propagation rate (Mao et al., 2023; Yang et al., 2017). Similar conclusions have been reported in recent studies showing that drought propagation is jointly controlled by climatic forcing and catchment characteristics, and that groundwater-related drought responses are generally slower and more persistent than streamflow responses (Geng et al., 2024).

Substantial inter-basin differences were also identified. Compared with the JRB, the TRB shows a significantly longer propagation time from meteorological drought to baseflow drought (Fig. 12d). This contrast suggests that the two basins differ not only in their overall hydrological conditions, but also in the way drought signals are transmitted through soil water, subsurface storage, and groundwater-supported discharge. As shown in Fig. 13a and b, the TRB has higher streamflow and baseflow levels than the JRB, implying a stronger capacity of catchment storage to buffer short-term precipitation deficits (Zhou et al., 2021). Under such conditions, the effect of precipitation deficits on baseflow can be buffered for a longer period, indicating that drought signals are not immediately transferred from atmospheric forcing to the groundwater-controlled component of river flow. In contrast, in the JRB, precipitation deficits can be transmitted more rapidly to soil moisture depletion and shallow subsurface storage reduction, which then accelerates the decline in groundwater-supported discharge and shortens the propagation time to baseflow drought. In addition, topographic differences may further contribute to this contrast (Wang et al., 2021a; Xiong et al., 2025). The steeper terrain of the JRB may favor faster runoff generation and reduce the buffering effect of near-surface storage, thereby allowing drought signals to propagate more rapidly through the hydrological system (Saft et al., 2016). By contrast, the relatively stronger buffering conditions in the TRB may delay the transmission of meteorological drought to baseflow drought. This interpretation is consistent with recent studies emphasizing that drought propagation and its spatial heterogeneity are strongly regulated by basin storage conditions, hydrological buffering, climatic variability, and catchment properties (Fang et al., 2020; Geng et al., 2024; Xiong et al., 2025).

4.2. Limitations and expectations

This study employed hydrological variables simulated by the improved SWAT model to analyze drought characteristics and propagation patterns, providing new insights into drought evolution processes. Nevertheless, several limitations remain. Drought propagation is influenced by multiple factors, including soil properties, vegetation dynamics, and human activities. In this study, the analysis mainly focused on natural meteorological and hydrological processes, while the effects of land use change, irrigation regulation, and other anthropogenic influences were not explicitly considered. In addition, the SWAT model was constructed using a static land use dataset. Although vegetation dynamics were partially represented using remotely sensed LAI and phenological data, large-scale vegetation restoration associated with the “Grain for Green” project on the Loess Plateau may still affect runoff generation and soil water retention, thereby influencing the effective curve number (CN) values during long-term simulations. Future studies could incorporate dynamic land use datasets or scenario-based simulations to better represent time-varying land surface conditions.

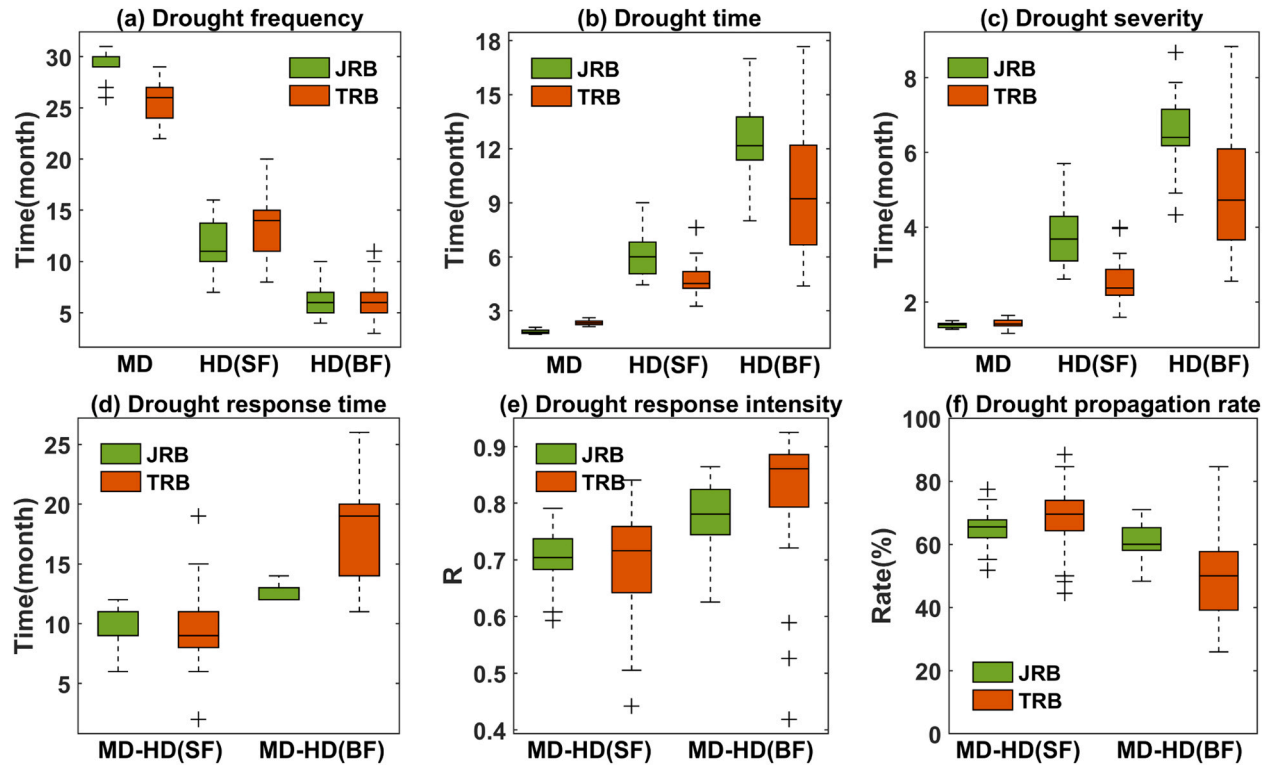


Fig. 12. Boxplot of drought characteristics, propagation time, intensity, and effective propagation rate. Note: MD, Meteorological Drought; HD, Hydrological Drought; SF, Streamflow; BF, Baseflow.

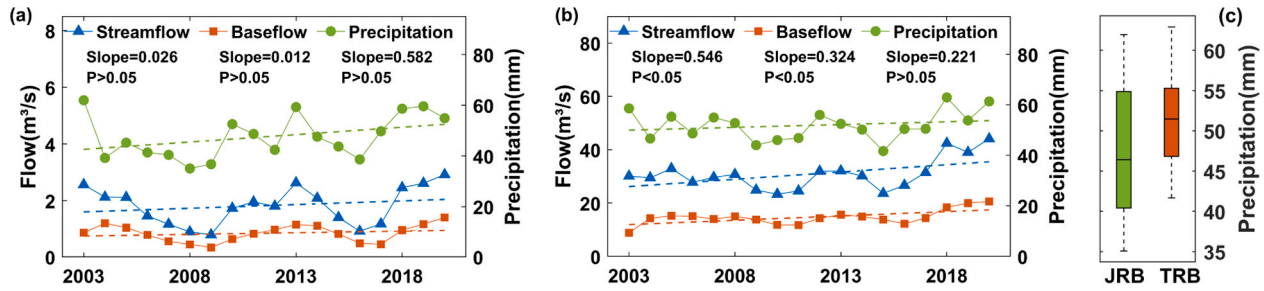


Fig. 13. Multi-year variation trends and differences in precipitation, streamflow, and baseflow in the JRB and TRB.

Moreover, uncertainties may arise from the selection of drought indices and their statistical distributions. Conventional drought indices and probability distribution functions may not fully capture basin-specific hydrological and climatic characteristics. Future research could explore a broader range of drought indicators and more flexible statistical approaches, particularly those accounting for nonlinear characteristics, to improve the reliability of drought identification and analysis.

In addition, parameter selection in the baseflow separation method may introduce uncertainties in estimating groundwater contributions to streamflow, which could affect the identification of hydrological drought events and the estimation of drought propagation characteristics. Future studies could further evaluate these uncertainties through multi-model comparisons or ensemble simulations to enhance the robustness of drought propagation assessments.

5. Conclusions

This study integrated an improved SWAT model with the Eckhardt digital filter and run theory to investigate the characteristics and propagation patterns of meteorological drought, streamflow drought, and baseflow drought in the Jing River Basin (JRB) and Tao River Basin (TRB). The results showed that meteorological droughts occurred more frequently than hydrological droughts, whereas hydrological droughts generally exhibited longer duration and greater severity, especially baseflow droughts. In both basins, the propagation from meteorological drought to streamflow drought was faster and more effective than that to baseflow drought, indicating a more direct response of streamflow to precipitation deficits and a stronger lag effect in baseflow. In addition, clear inter-basin differences were identified, with the TRB showing a longer propagation time from meteorological drought to baseflow drought than the JRB, reflecting the important role of basin hydrological characteristics in regulating drought evolution.

This study has several limitations that should be acknowledged. The hydrological simulations were conducted using a static land use dataset, which may not fully capture the long-term effects of land use change and vegetation restoration on hydrological processes and drought propagation, especially in the Loess Plateau under the “Grain for Green” project. As a result, the influences of changing land cover on runoff generation, soil water retention, and groundwater recharge may be underestimated. Future research should incorporate dynamic land use data or scenario-based land use inputs to better represent temporal changes in land surface conditions and to improve the reliability of drought propagation analysis.

The findings of this study provide useful implications for drought monitoring and water resources management in ecologically fragile regions. In particular, the results highlight the importance of baseflow in drought evolution and may provide a reference for hydrological drought assessment and drought mitigation in other climate-sensitive or data-scarce basins.

CRediT authorship contribution statement

Lei Jin: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Huazhu Xue:** Writing – review & editing, Supervision. **Guangyu Ma:** Supervision. **Yinglan A.:** Writing – review & editing, Supervision, Funding acquisition. **Guoqiang Wang:** Funding acquisition, Supervision. **Kaiji Li:** Writing – review & editing. **Jin Wu:** Writing – review & editing. **Yuntao Wang:** Writing – review & editing. **Baolin Xue:** Writing – review & editing.

Declaration of Competing Interest

The author(s) declared no potential conflicts of interest with respect to the research, author-ship, and/or publication of this article.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ejrh.2026.103538](https://doi.org/10.1016/j.ejrh.2026.103538).

Data availability

Data will be made available on request.

References

- Aloui, S., Mazzoni, A., Elomri, A., Aouissi, J., Boufekane, A., Zghibi, A., 2023. A review of Soil and Water Assessment Tool (SWAT) studies of Mediterranean catchments: applications, feasibility, and future directions. *J. Environ. Manag.* 326, 116799.

- Barker, L.J., Hannaford, J., Chiverton, A., Svensson, C., 2016. From meteorological to hydrological drought using standardised indicators. *Hydrol. Earth Syst. Sci.* 20 (6), 2483–2505.
- Bawa, A., Mendoza, K., Srinivasan, R., Parmar, R., Smith, D., Wolfe, K., Johnston, J.M., Corona, J., 2024. Calibration using R-programming and parallel processing at the HUC12 subbasin scale in the Mid-Atlantic region: development of national SWAT hydrologic calibration. *Environ. Model. Softw.* 176, 106019.
- Carroll, R.W.H., Manning, A.H., Williams, K.H., 2025. The role of bedrock circulation depth and porosity in mountain streamflow response to prolonged drought. *Geophys. Res. Lett.* 52 (4) e2024GL112927.
- Chen, J., Jönsson, P., Tamura, M., Gu, Z., Matsushita, B., Eklundh, L., 2004. A simple method for reconstructing a high-quality NDVI time-series data set based on the Savitzky-Golay filter. *Remote Sens. Environ.* 91 (3), 332–344.
- Chen, J., Fan, Y., Zhang, Y., Peng, J., Zhang, J., Cao, C., 2024. Comprehensive propagation characteristics between paired meteorological and hydrological drought events: insights from various underlying surfaces. *Atmos. Res.* 299, 107193.
- Chen, P., Zhang, X., Li, E., Han, S., 2025. Analyzing the impact of land use changes on hydrological drought in the Yiluo river basin based on SWAT. *Environ. Sustain. Indic.* 28, 101052.
- Chen, S., Fu, Y.H., Wu, Z., Hao, F., Hao, Z., Guo, Y., Geng, X., Li, X., Zhang, X., Tang, J., Singh, V.P., Zhang, X., 2023. Informing the SWAT model with remote sensing detected vegetation phenology for improved modeling of ecohydrological processes. *J. Hydrol.* 616, 128817.
- Dash, S.S., Sahoo, B., Raghuvanshi, N.S., 2019. A SWAT-Copula based approach for monitoring and assessment of drought propagation in an irrigation command. *Ecol. Eng.* 127, 417–430.
- Ding, Y., Xu, J., Wang, X., Cai, H., Zhou, Z., Sun, Y., Shi, H., 2021b. Propagation of meteorological to hydrological drought for different climate regions in China. *J. Environ. Manag.* 283, 111980.
- Ding, Y., Gong, X., Xing, Z., Cai, H., Zhou, Z., Zhang, D., Sun, P., Shi, H., 2021a. Attribution of meteorological, hydrological and agricultural drought propagation in different climatic regions of China. *Agric. Water Manag.* 255, 106996.
- Du, H., Tan, M.L., Samat, N., Chen, Z., Zhang, F., 2025. Integrating a comprehensive index and the SWAT+ model to assess drought characteristics and risks under solar radiation modification. *Ecol. Indic.* 178, 113852.
- Eckhardt, K., 2008. A comparison of baseflow indices, which were calculated with seven different baseflow separation methods. *J. Hydrol.* 352 (1), 168–173.
- Fang, W., Huang, S., Huang, Q., Huang, G., Wang, H., Leng, G., Wang, L., 2020. Identifying drought propagation by simultaneously considering linear and nonlinear dependence in the Wei River basin of the Loess Plateau, China. *J. Hydrol.* 591, 125287.
- Geng, G., Zhang, B., Gu, Q., He, Z., Zheng, R., 2024. Drought propagation characteristics across China: time, probability, and threshold. *J. Hydrol.* 631, 130805.
- Gevaert, A.I., Veldkamp, T.I.E., Ward, P.J., 2018. The effect of climate type on timescales of drought propagation in an ensemble of global hydrological models. *Hydrol. Earth Syst. Sci.* 22 (9), 4649–4665.
- Ghaneai, P., Foroumandi, E., Stahl, K., Ajami, H., Wanders, N., Moradkhani, H., 2026. Four decades of baseflow drought analysis reveals varying contributions of climatic drivers and physical controls. *Earth's Future* 14 (1), e2025EF006934.
- Gu, L., Chen, J., Yin, J., Xu, C.-Y., Chen, H., 2020. Drought hazard transferability from meteorological to hydrological propagation. *J. Hydrol.* 585, 124761.
- Guo, Y., Huang, S., Huang, Q., Leng, G., Fang, W., Wang, L., Wang, H., 2020. Propagation thresholds of meteorological drought for triggering hydrological drought at various levels. *Sci. Total Environ.* 712, 136502.
- Han, Z., Huang, Q., Huang, S., Leng, G., Bai, Q., Liang, H., Wang, L., Zhao, J., Fang, W., 2021. Spatial-temporal dynamics of agricultural drought in the Loess Plateau under a changing environment: characteristics and potential influencing factors. *Agric. Water Manag.* 244, 106540.
- Han, Z., Huang, S., Zhao, J., Leng, G., Huang, Q., Zhang, H., Li, Z., 2023. Long-chain propagation pathways from meteorological to hydrological, agricultural and groundwater drought and their dynamics in China. *J. Hydrol.* 625, 130131.
- Huang, M., Gallichand, J., Wang, Z., Goulet, M., 2006. A modification to the soil conservation service curve number method for steep slopes in the Loess Plateau of China. *Hydrol. Process.* 20 (3), 579–589.
- Huang, S., Li, P., Huang, Q., Leng, G., Hou, B., Ma, L., 2017. The propagation from meteorological to hydrological drought and its potential influence factors. *J. Hydrol.* 547, 184–195.
- Huang, X., Yang, X., Wu, F., Zhang, J., 2025. Future propagation characteristics of meteorological drought to hydrological drought in the Yellow River basin. *J. Hydrol.* 649, 132443.
- Jia, X., Zhu, P., Wei, X., Zhu, Y., Huang, M., Hu, W., Wang, Y., Turkeltaub, T., Binley, A., Horton, R., Shao, Ma, 2024. Bringing ancient loess critical zones into a new era of sustainable development goals. *Earth Sci. Rev.* 255, 104852.
- Jiang, A., Zhang, W., Liu, X., Zhou, F., Li, A., Peng, H., Wang, H., 2024. Improving hydrological process simulation in mountain watersheds: integrating WRF model gridded precipitation data into the SWAT model. *J. Hydrol.* 639, 131687.
- Jiang, C., Zhang, H., Tang, Z., Labzovskii, L., 2017. Evaluating the coupling effects of climate variability and vegetation restoration on ecosystems of the Loess Plateau, China. *Land Use Policy* 69, 134–148.
- Jin, L., Xue, H., Dong, G., Han, Y., Li, Z., Lian, Y., 2024. Coupling the remote sensing data-enhanced SWAT model with the bidirectional long short-term memory model to improve daily streamflow simulations. *J. Hydrol.* 634, 131117.
- Kang, H., Sridhar, V., 2017. Combined statistical and spatially distributed hydrological model for evaluating future drought indices in Virginia. *J. Hydrol. Reg. Stud.* 12, 253–272.
- Lai, C., Zhong, R., Wang, Z., Wu, X., Chen, X., Wang, P., Lian, Y., 2019. Monitoring hydrological drought using long-term satellite-based precipitation data. *Sci. Total Environ.* 649, 1198–1208.
- Lai, G., Luo, J., Li, Q., Qiu, L., Pan, R., Zeng, X., Zhang, L., Yi, F., 2020. Modification and validation of the SWAT model based on multi-plant growth mode, a case study of the Meijiang River Basin, China. *J. Hydrol.* 585, 124778.
- Lee, S., Ajami, H., 2023. Comprehensive assessment of baseflow responses to long-term meteorological droughts across the United States. *J. Hydrol.* 626, 130256.
- Lin, Q., Wu, Z., Zhang, Y., Peng, T., Chang, W., Guo, J., 2023. Propagation from meteorological to hydrological drought and its application to drought prediction in the Xijiang River basin, South China. *J. Hydrol.* 617, 128889.
- Liu, D., 2020. A rational performance criterion for hydrological model. *J. Hydrol.* 590, 125488.
- Liu, Y., Wang, W., Zhang, J., Li, Z., 2024. Characteristics and sources of chemical composition in precipitation on the Loess Plateau of China. *Sci. Total Environ.* 935, 173401.
- Mao, R., Shi, A., Song, J., Xu, W., Tang, B., Li, B., 2023. Response of the runoff process to meteorological drought: Baseflow index as an important indicator. *J. Environ. Manag.* 345, 118843.
- Miller, M.P., Buto, S.G., Susong, D.D., Rumsey, C.A., 2016. The importance of base flow in sustaining surface water flow in the Upper Colorado River Basin. *Water Resour. Res.* 52 (5), 3547–3562.
- Mishra, A.K., Singh, V.P., 2010. A review of drought concepts. *J. Hydrol.* 391 (1), 202–216.
- Moriasi, D.N., W. Gitau, M., Pai, N., Dagupati, P., 2015. Hydrologic and water quality models: performance measures and evaluation criteria. *Trans. ASABE* 58 (6), 1763–1785.
- Pan, Y., Zhu, Y., Lü, H., Yagci, A.L., Fu, X., Liu, E., Xu, H., Ding, Z., Liu, R., 2023. Accuracy of agricultural drought indices and analysis of agricultural drought characteristics in China between 2000 and 2019. *Agric. Water Manag.* 283, 108305.
- Pan, Z., Liu, P., Xu, C.Y., Cheng, L., Tian, J., Cheng, S., Xie, K., 2020. The influence of a prolonged meteorological drought on catchment water storage capacity: a hydrological-model perspective. *Hydrol. Earth Syst. Sci.* 24 (9), 4369–4387.
- Pang, S., Wang, X., Melching, C.S., Feger, K.-H., 2020. Development and testing of a modified SWAT model based on slope condition and precipitation intensity. *J. Hydrol.* 588, 125098.
- Qiu, D., Xu, R., Gao, P., Mu, X., 2024. Effect of vegetation restoration type and topography on soil water storage and infiltration capacity in the Loess Plateau, China. *CATENA* 241, 108079.

- Saft, M., Peel, M.C., Western, A.W., Zhang, L., 2016. Predicting shifts in rainfall-runoff partitioning during multiyear drought: Roles of dry period and catchment characteristics. *Water Resour. Res.* 52 (12), 9290–9305.
- Shi, H., Zhou, Z., Liu, L., Liu, S., 2022. A global perspective on propagation from meteorological drought to hydrological drought during 1902–2014. *Atmos. Res.* 280, 106441.
- Wan, L., Bento, V.A., Qu, Y., Qiu, J., Song, H., Zhang, R., Wu, X., Xu, F., Lu, J., Wang, Q., 2023. Drought characteristics and dominant factors across China: insights from high-resolution daily SPEI dataset between 1979 and 2018. *Sci. Total Environ.* 901, 166362.
- Wang, C., Gomez-Velez, J.D., Wilson, J.L., 2022. Dynamic coevolution of baseflow and multiscale groundwater flow system during prolonged droughts. *J. Hydrol.* 609, 127657.
- Wang, H., Zhao, H., Wang, F., Yan, B., Tang, L., Du, Y., Cui, L., 2024. Study on the multi-type drought propagation process and driving factors on the Tibetan Plateau. *J. Hydrol.* 645, 132162.
- Wang, J., Wang, W., Cheng, H., Wang, H. and Zhu, Y. 2021a Propagation from Meteorological to Hydrological Drought and Its Influencing Factors in the Huaihe River Basin, p. 1985.
- Wang, S., Huang, S., Wang, C., Zhang, X., Wu, J., Gulakhmadov, A., Niyogi, D., Chen, N., 2025. Global anthropogenic effects on meteorological—hydrological—soil moisture drought propagation: Historical analysis and future projection. *J. Hydrol.* 653, 132755.
- Wang, T., Tu, X., Singh, V.P., Chen, X., Lin, K., 2021b. Global data assessment and analysis of drought characteristics based on CMIP6. *J. Hydrol.* 596, 126091.
- Wang, Y., Wang, S., Chen, Y., Wang, F., Liu, Y., Zhao, W., 2023. Anthropogenic drought in the Yellow River basin: Multifaceted and weakening connections between meteorological and hydrological droughts. *J. Hydrol.* 619, 129273.
- Wu, D., Chen, J., Xiong, L., Lee, J.-H., Kim, J.-S., Moon, H.-T., 2024a. Assessing global drought conditions under climate change: a comparison of stationary and non-stationary approaches and identification of hotspot regions. *J. Hydrol.* 640, 131663.
- Wu, J., Gao, X.J., 2013. A gridded daily observation dataset over China region and comparison with the other datasets. *Chin. J. Geophys.* 56 (4), 1102–1111.
- Wu, J., Chen, X., Yao, H., Gao, L., Chen, Y., Liu, M., 2017. Non-linear relationship of hydrological drought responding to meteorological drought and impact of a large reservoir. *J. Hydrol.* 551, 495–507.
- Wu, J., Liu, Z., Yao, H., Chen, X., Chen, X., Zheng, Y., He, Y., 2018. Impacts of reservoir operations on multi-scale correlations between hydrological drought and meteorological drought. *J. Hydrol.* 563, 726–736.
- Wu, Q., Jiang, X., Shi, X., Zhang, Y., Liu, Y., Cai, W., 2024b. Spatiotemporal evolution characteristics of soil erosion and its driving mechanisms - a case Study: loess Plateau, China. *CATENA* 242, 108075.
- Xiong, H., Han, J., Wang, Y., 2025. Propagation from meteorological to hydrological drought: characteristics and influencing factors. *Water Resour. Res.* 61 (4) e2024WR037765.
- Xu, J., Zou, L., Xia, J., Zhang, Y., Wang, F., Chen, X., 2025. Investigating the effects of climate and vegetation changes on spatiotemporal variation of baseflow in the Yellow River Basin. *J. Hydrol.* 651, 132517.
- Xue, C., Ghirardelli, A., Chen, J., Tarolli, P., 2024. Investigating agricultural drought in Northern Italy through explainable machine learning: insights from the 2022 drought. *Comput. Electron. Agric.* 227, 109572.
- Yang, X., Wu, F., Yuan, S., Ren, L., Sheffield, J., Fang, X., Jiang, S., Liu, Y., 2024. Quantifying the impact of human activities on hydrological drought and drought propagation in China using the PCR-GLOBWB v2.0 model. *Water Resour. Res.* 60 (1) e2023WR035443.
- Yang, Y., McVicar, T.R., Donohue, R.J., Zhang, Y., Roderick, M.L., Chiew, F.H.S., Zhang, L., Zhang, J., 2017. Lags in hydrologic recovery following an extreme drought: assessing the roles of climate and catchment characteristics. *Water Resour. Res.* 53 (6), 4821–4837.
- Yao, Y., Zheng, C., Andrews, C., Zheng, Y., Zhang, A., Liu, J., 2017. What controls the partitioning between baseflow and mountain block recharge in the Qinghai-Tibet Plateau? *Geophys. Res. Lett.* 44 (16), 8352–8358.
- Yu, J., Zou, L., Xia, J., Dou, M., Liu, H., Zuo, L., 2023. Future changes in hydrological drought across the Yangtze River Basin: identification, spatial-temporal characteristics, and concurrent probability. *J. Hydrol.* 625, 130057.
- Zhang, Q., Miao, C., Gou, J., Wu, J., Jiao, W., Song, Y., Xu, D., 2022. Spatiotemporal characteristics of meteorological to hydrological drought propagation under natural conditions in China. *Weather Clim. Extrem.* 38, 100505.
- Zhao, J., Zhang, N., Liu, Z., Zhang, Q., Shang, C., 2024. SWAT model applications: from hydrological processes to ecosystem services. *Sci. Total Environ.* 931, 172605.
- Zhou, Z., Shi, H., Fu, Q., Ding, Y., Li, T., Liu, S., 2021. Investigating the propagation from meteorological to hydrological drought by introducing the nonlinear dependence with directed information transfer index. *Water Resour. Res.* 57 (8) e2021WR030028.
- Zhou, Z., Wang, P., Li, L., Fu, Q., Ding, Y., Chen, P., Xue, P., Wang, T., Shi, H., 2024. Recent development on drought propagation: a comprehensive review. *J. Hydrol.* 645, 132196.
- Zhu, Y., Liu, Y., Wang, W., Singh, V.P., Ren, L., 2021. A global perspective on the probability of propagation of drought: from meteorological to soil moisture. *J. Hydrol.* 603, 126907.
- Zou, R., Wang, X., Yin, Y., Ma, X., Yang, X., Huang, P., Ullah, I., 2024. Comparing spatio-temporal propagation patterns of hydrological and meteorological droughts: insights from SWAT modelling in the poyang lake basin. *CATENA* 243, 108183.